Stage 1

Title of the project :

Design and apply context-aware access controls based on user identity.

Overview :

Context-aware access control is a security approach that allows or denies access to resources based not only on the user's identity but also on the context of the access request. The context may include factors such as the user's location, device type, time of access, network condition, and behavior. This method improves security by ensuring that users can access systems only when their identity and surrounding conditions are verified and considered safe. The design of context-aware access control begins with user identity verification. Every user must have a unique identity, which is authenticated using methods such as passwords, biometric verification, or multi-factor authentication.

After authentication, the system evaluates the context information, such as whether the user is logging in from a trusted device, the geographic location, the time of the request, and the type of network being used. Once the context is analyzed, the system applies access control policies. These policies define what resources a user can access under specific conditions. For example, an employee may be allowed to access company data only during working hours or only from an authorized device. If the context does not meet the defined policy rules, the system may restrict access, request additional authentication, or block the request entirely.

Implementing context-aware access control improves system security, flexibility, and user management. It reduces the risk of unauthorized access and helps organizations protect sensitive information while still allowing legitimate users to work efficiently. This approach is widely used in modern cloud systems, enterprise networks, and Internet of Things (IoT) environments.

List of Teammates:

S.NO	NAME	COLLEGE	CONTACT
1.	M.AKSHAYA	HINDUSTHAN COLLEGE OF SCIENCE AND COMMERCE,ERODE	akshayamahesh06@gmail.com
2	P.JAYALALITHA	HINDUSTHAN COLLEGE OF SCIENCE AND COMMERCE,ERODE	jayalalitha250749@gmail.com
3	B.DHIVYA	HINDUSTHAN COLLEGE OF SCIENCE AND COMMERCE,ERODE	bdhivya424@gmail.com
4	V.S SRIYAMUNA	HINDUSTHAN COLLEGE OF SCIENCE AND COMMERCE,ERODE	yamunavasanthi368@gmail.com
5	N.MAHIMA	HINDUSTHAN COLLEGE OF SCIENCE AND COMMERCE,ERODE	mahimanagaraj1866@gamil.com

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Report:

1.Vulnerability name: Cross site scripting (stored)

CWE: 79

OWASP category: A03:2021 -Injections

Description:

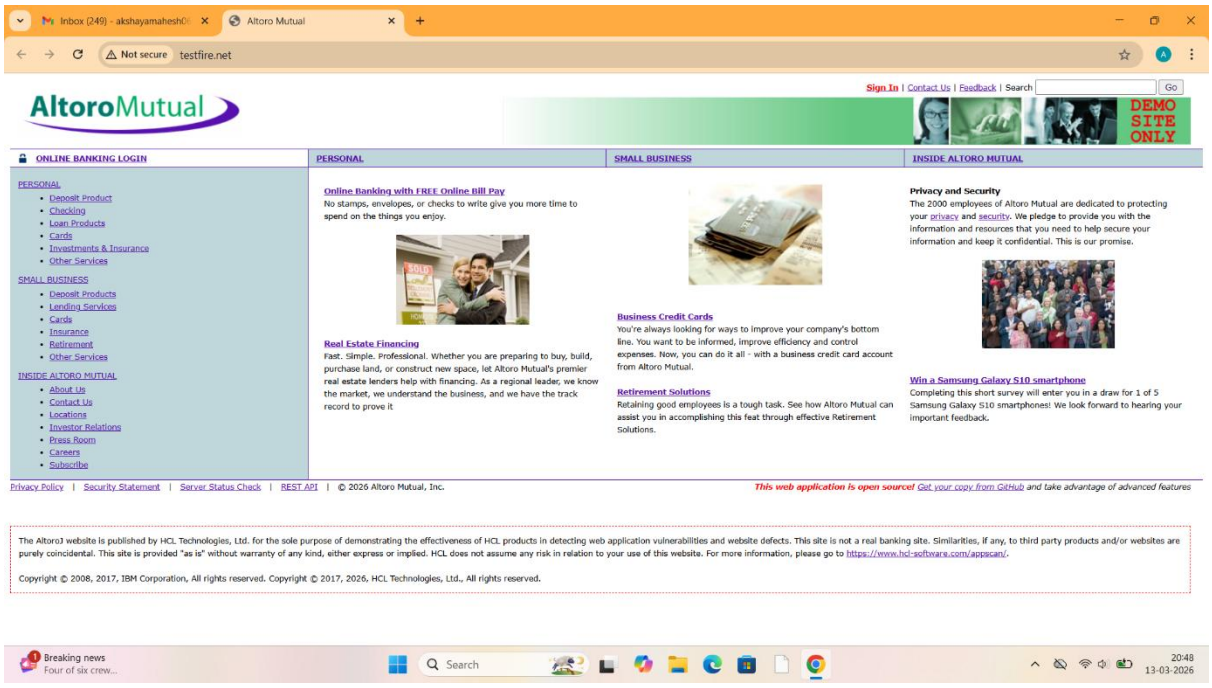
It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

Business Impact:

The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area that is displayed to either many users or particularly interesting users.

Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

Accessing the URL:



Input code to perform the vulnerability:

Displaying the harmless pop up alert box says"hacking":

2.Vulnerability name: Broken access control

CWE: 285

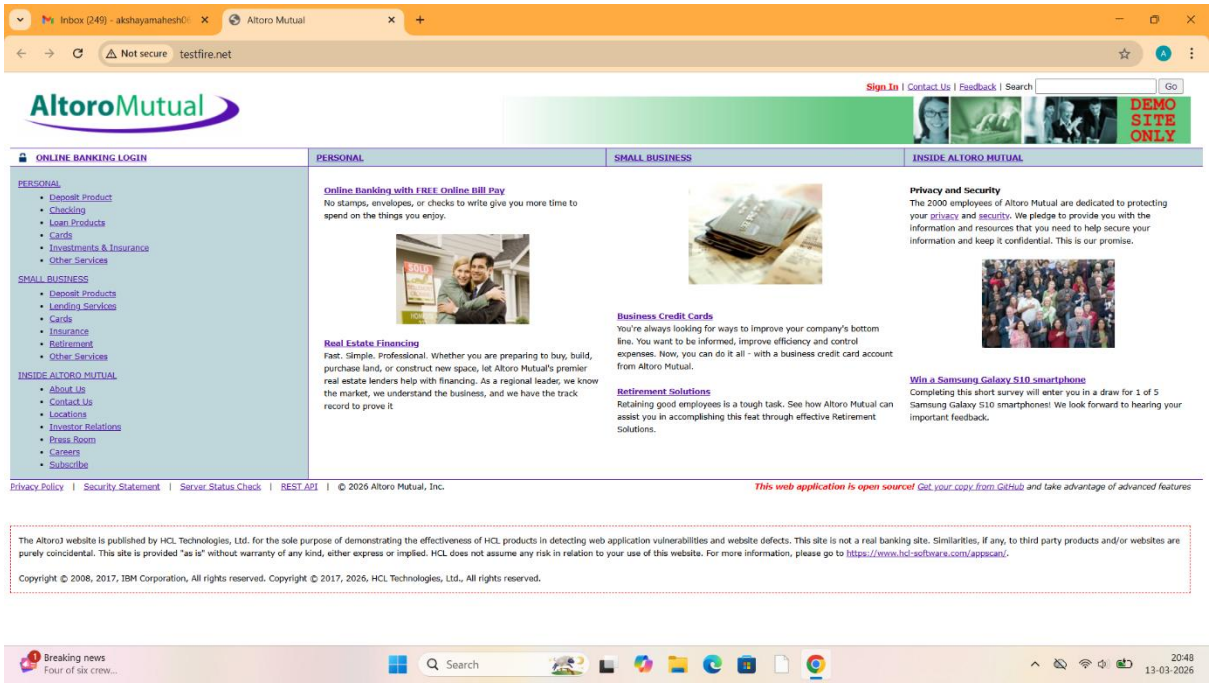
Description:

When access control checks are not applied consistently - or not at all - users are able to access data or perform actions that they should not be allowed to perform. This can lead to a wide range of problems, including information exposures, denial of service, and arbitrary code execution.

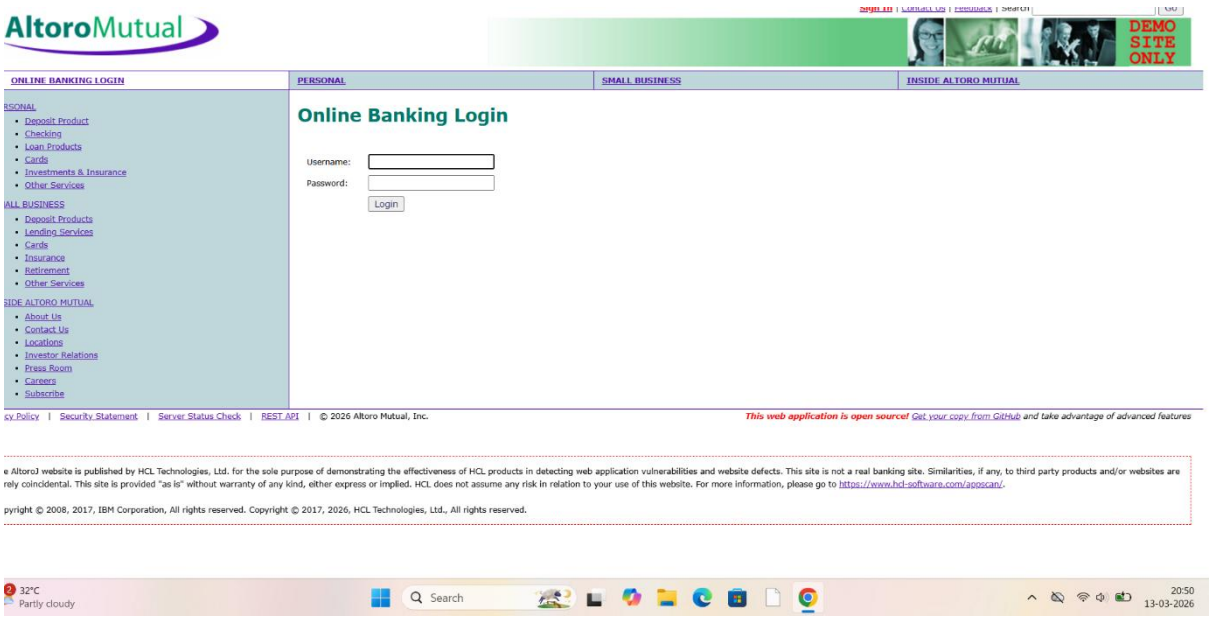
Business Impact:

Wide-ranging business impact, including financial losses, reputation damage, legal consequences, operational disruption, and loss of customer trust. Addressing this vulnerability is crucial to protect the organization and its stakeholders from these potential negative effects.

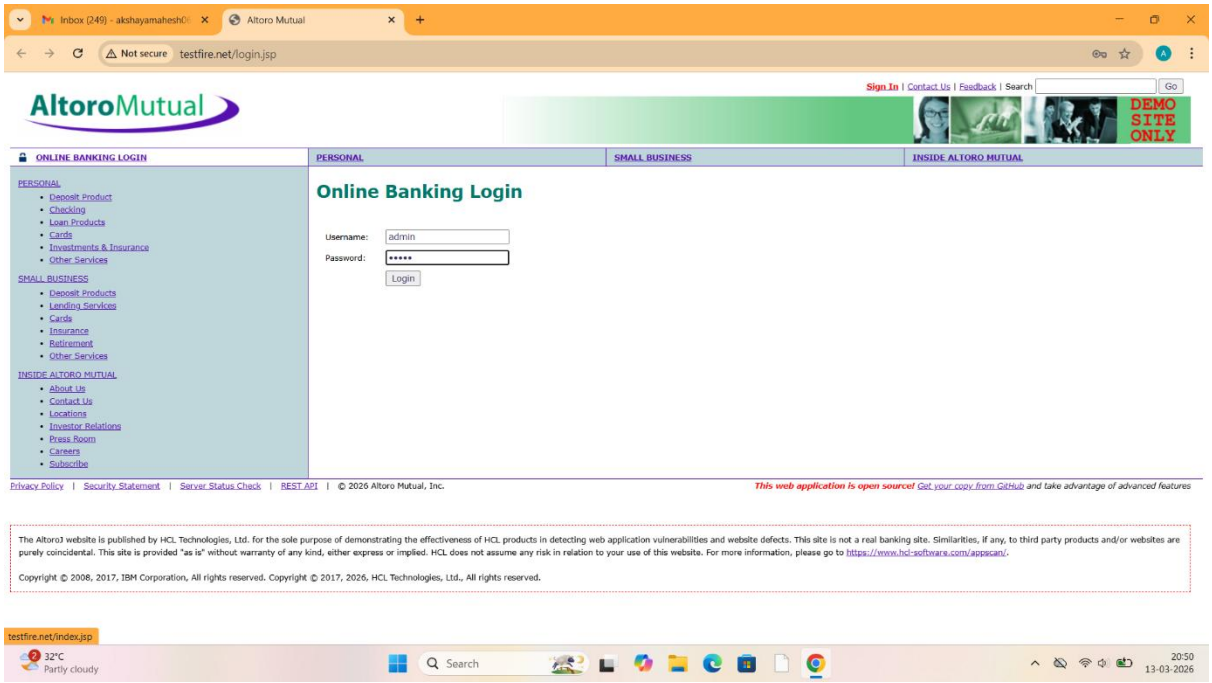
Accessing the URL:



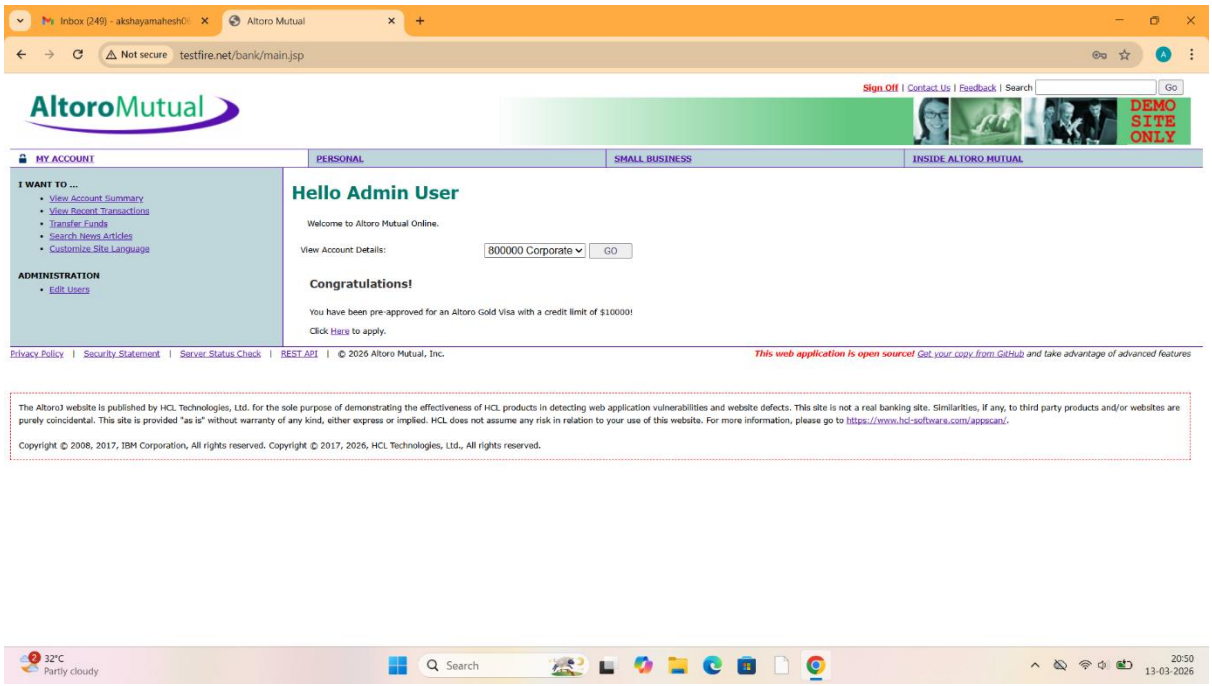
Logging in:



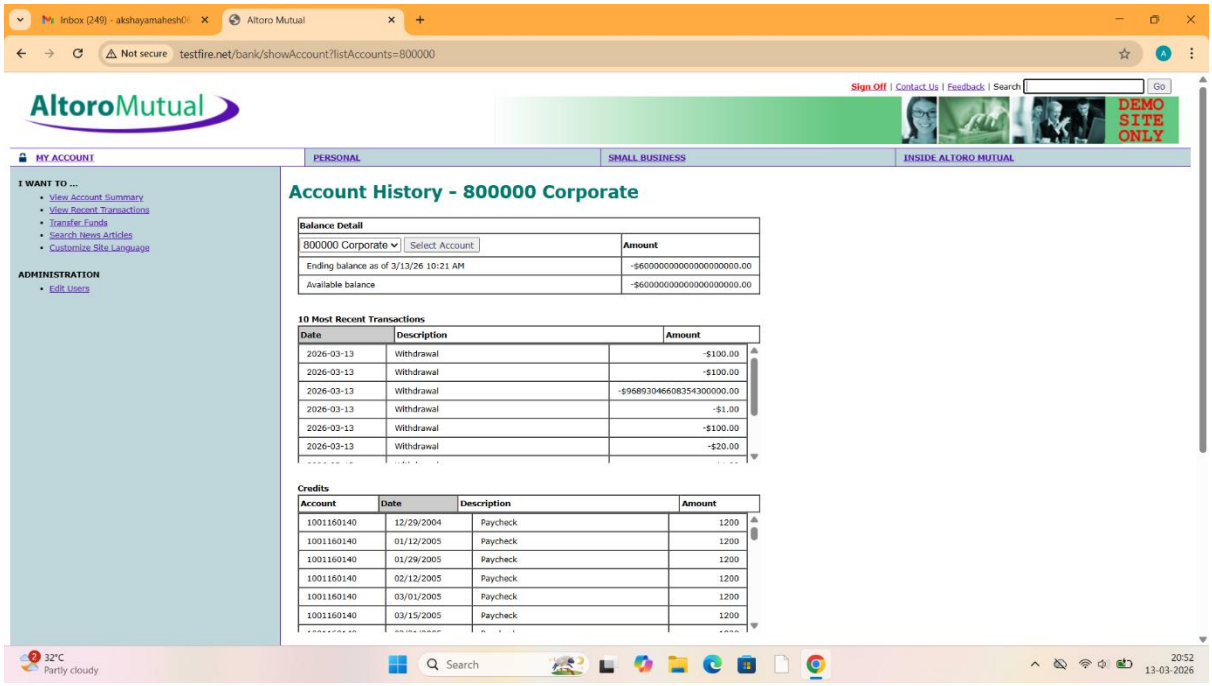
Entering Details:



Redirecting to Admin User:



Displaying the Details:



Editing the User Details and Deleting the logins:

Inbox (249) - akshayamahesh0 | Altoro Mutual

Not secure testfire.net/admin/admin.jsp

Sign Off | Contact Us | Feedback | Search Go

AltoroMutual

DEMO SITE ONLY

MY ACCOUNT | PERSONAL | SMALL BUSINESS | INSIDE ALTORO MUTUAL

I WANT TO ...
- View Account Summary
- View Recent Transactions
- Transfer Funds
- Search News Articles
- Customize Site Language

ADMINISTRATION
- Edit Users

Edit User Information

Add an account to an existing user

Users: admin

Account Types: Savings

Add Account

Change user's password

Users: admin

Password:

Confirm:

Change Password

Add an new user

First Name:
Last Name:

Username:

Password:
Confirm:

Add User

It is highly recommended that you leave the username as first initial last name.

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3.Vulnerability name: SQL injection

CWE: 89

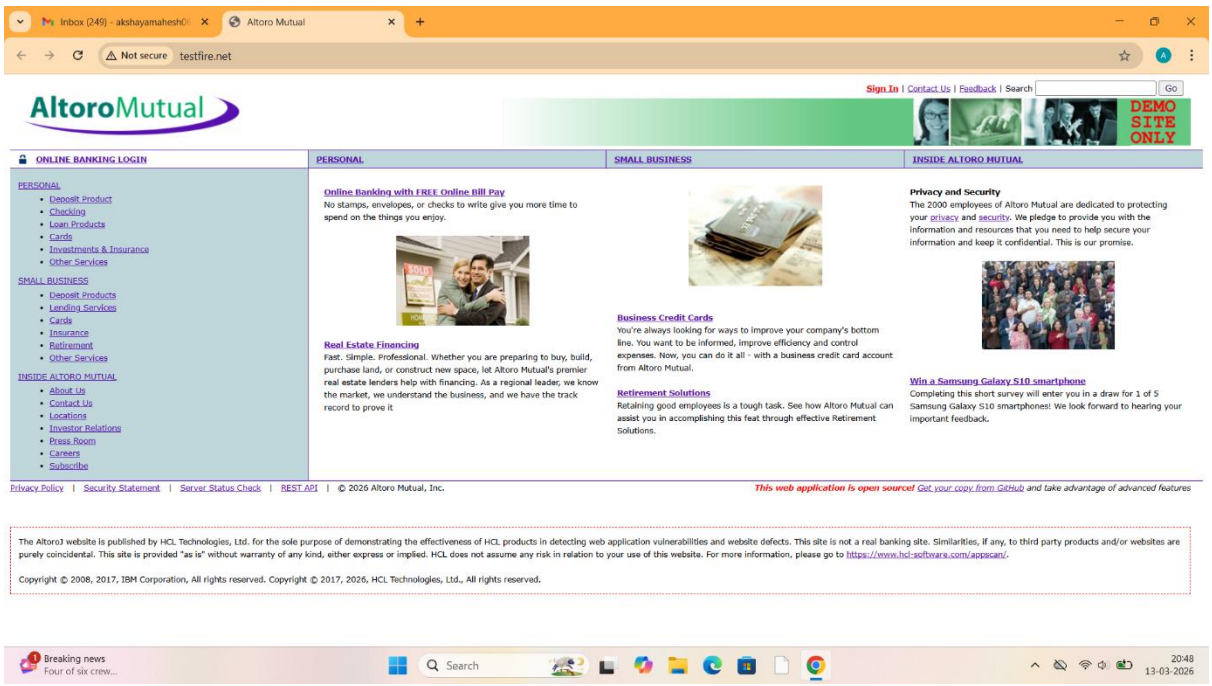
Description:

The product constructs all or part of an SQL command using externally-influenced input from an upstream component, but it does not neutralize or incorrectly neutralizes special elements that could modify the intended SQL command when it is sent to a downstream component.

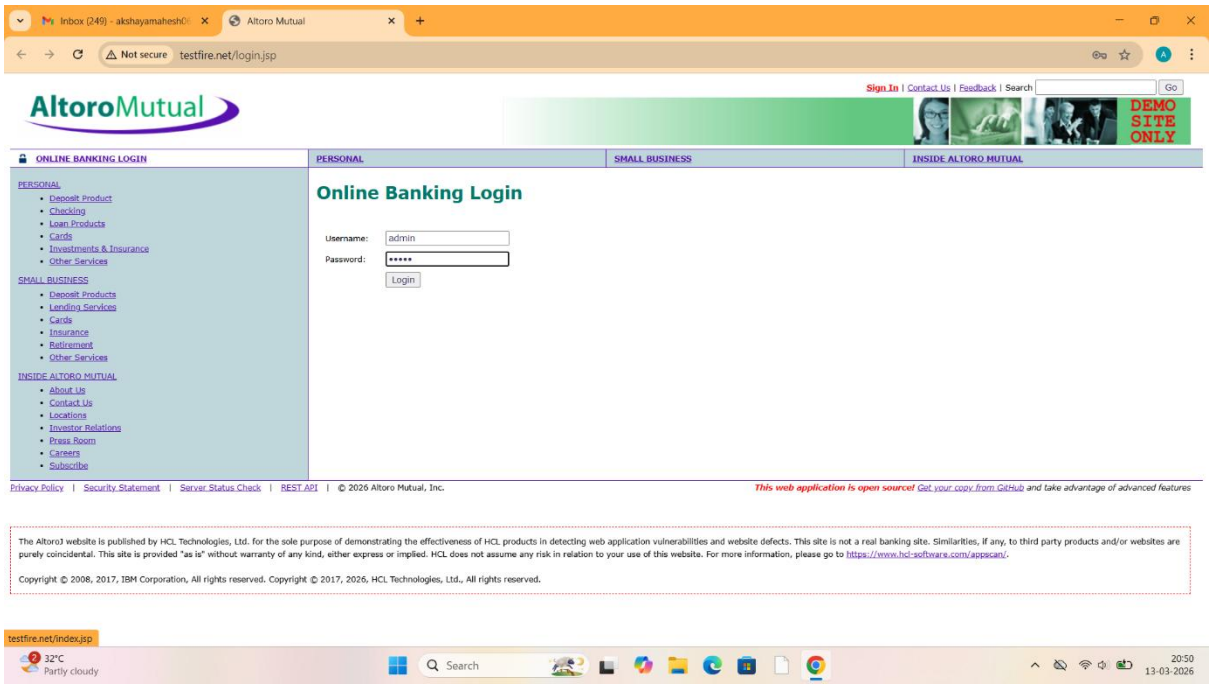
Business Impact:

In summary, it is crucial to underscore that CWE-89, known as SQL Injection, can exert a profound and diverse business impact. This encompasses critical facets such as data breaches, financial setbacks, harm to reputation, legal ramifications, and operational turmoil. Hence, the imperative of preventing and remedying SQL injection vulnerabilities cannot be overstated, as it is indispensable for fortifying the security and continuity of an organization's applications and data.

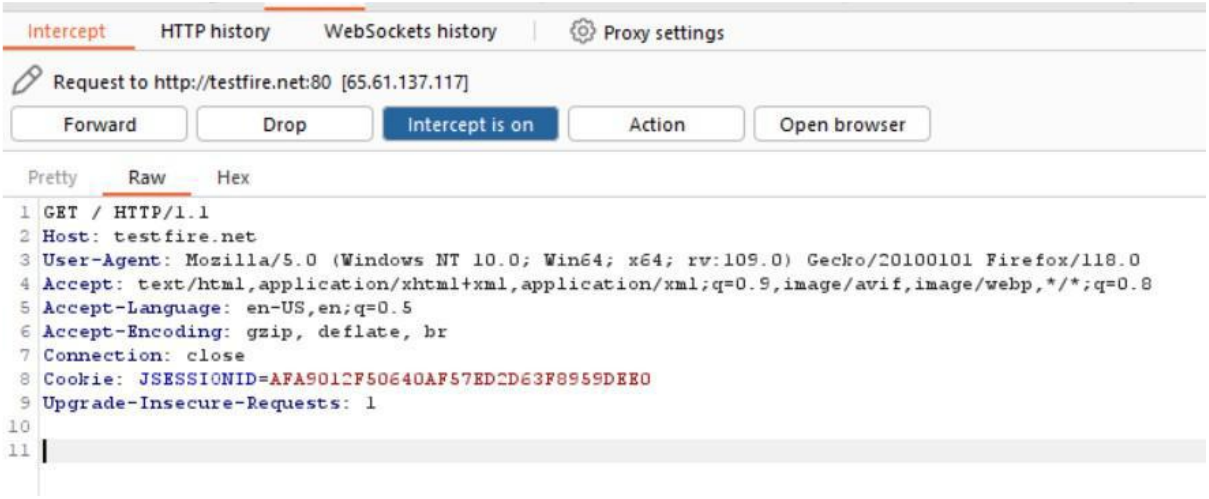
Accessing the URL:



Sign in to the admin privileges:



Using burp suite to get request from the website:



Dashboard | Target | Proxy | Intruder | Repeater | Collaborator | Sequencer | Decoder | Comparer | Logger | Organizer | Extensions | Learn

Intercept | HTTP history | WebSockets history | Proxy settings

Request to http://testfire.net:80 [65.61.137.117]

Forward | Drop | Intercept is on | Action | Open browser

Pretty | Raw | Hex

```
1  POST /doLogin HTTP/1.1
2  Host: testfire.net
3  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/118.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Content-Type: application/x-www-form-urlencoded
8  Content-Length: 45
9  Origin: http://testfire.net
10 Connection: close
11 Referer: http://testfire.net/login.jsp
12 Cookie: JSESSIONID=AFA9012F50640AF57ED2D63F8959DEE0
13 Upgrade-Insecure-Requests: 1
14
15 uid=%27+or+1%3D1--&passw=1234&btnSubmit=Login
```

Forward request multiple times to proceed to the login page :

Inbox (249) - akshayamahesh0 | Altoro Mutual

Not secure testfire.net/bank/main.jsp

AltoroMutual

Sign Off | Contact Us | Feedback | Search Go

DEMO SITE ONLY

MY ACCOUNT | PERSONAL | SMALL BUSINESS | INSIDE ALTORO MUTUAL

I WANT TO ...
- View Account Summary
- View Recent Transactions
- Transfer Funds
- Search News Articles
- Customize Site Language

ADMINISTRATION
- Edit Users

Hello Admin User

Welcome to Altoro Mutual Online.

View Account Details: 800000 Corporate GO

Congratulations!

You have been pre-approved for an Altoro Gold Visa with a credit limit of $10000!

Click Here to apply.

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We can know that sql injection worked and we got the admin privileges.

4.Vulnerability name: Broken authentication

CWE: 285

Description:

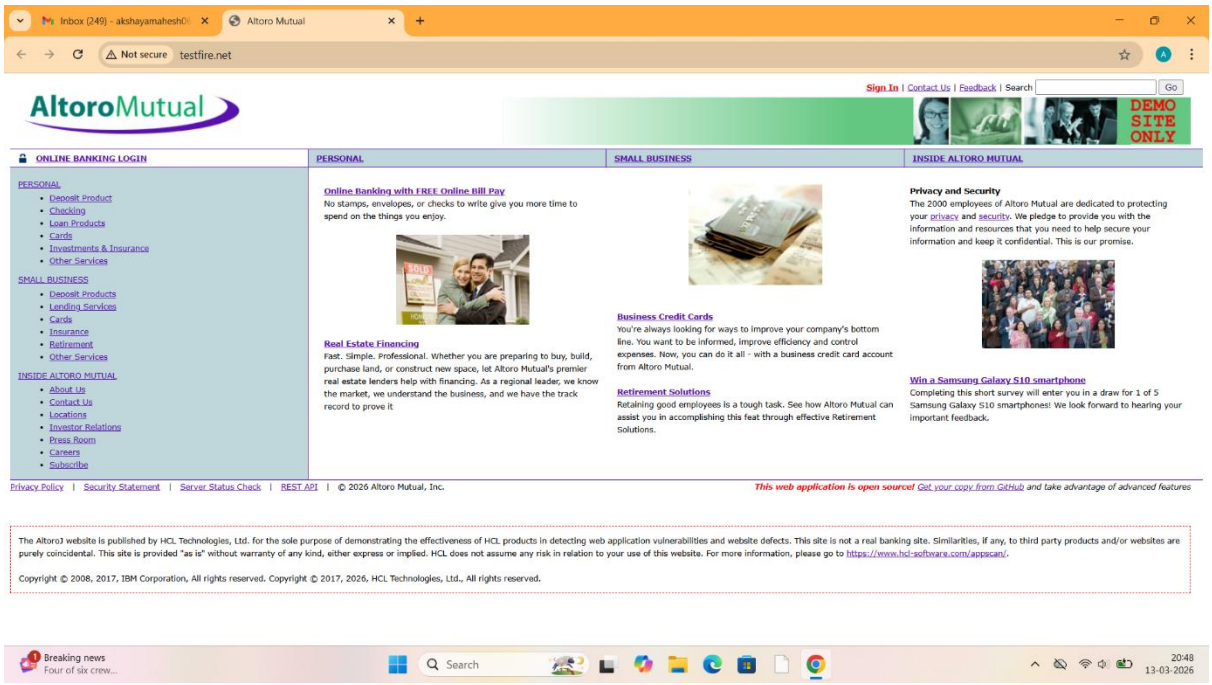
The system's authorization functionality does not prevent one user from gaining access to another user's data or record by modifying the key value identifying the data.

Business Impact:

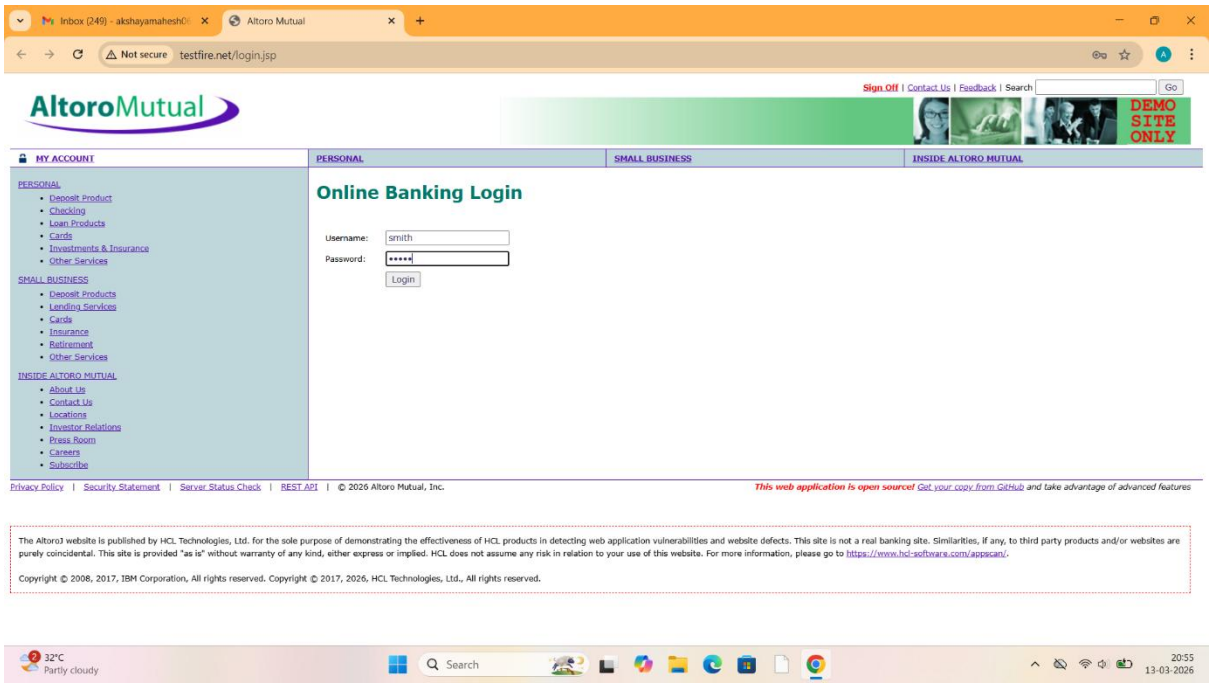
To effectively mitigate the business impact of CWE-285, it is imperative that organizations place a strong emphasis on bolstering their authentication and session management practices.

This should encompass the adoption of multi-factor authentication, the secure storage of credentials, meticulous session handling, and a commitment to conducting routine security assessments and testing. The rectification of these vulnerabilities stands as a critical imperative, safeguarding sensitive data, user identities, and the organization's overarching security stature and reputation.

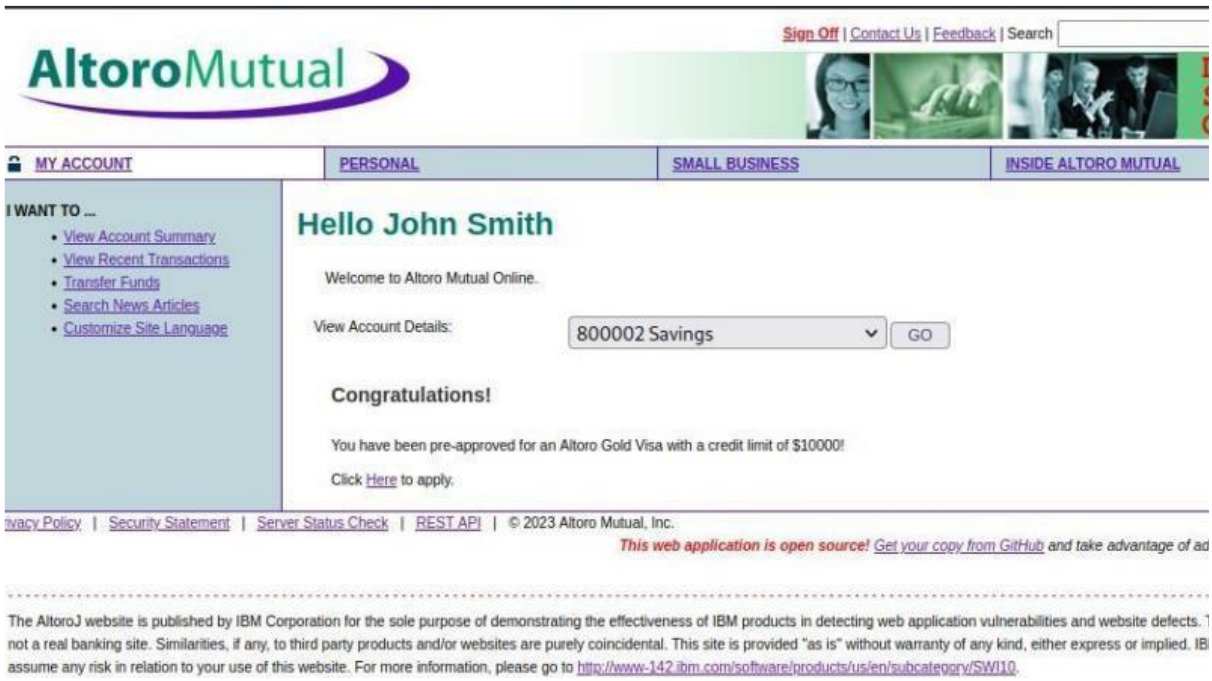
Accessing the URL:



Using different Logins:



We can directly access a particular user by simply knowing their username; we will add some characters after his user name as a sql injection to simply bypass the password.



This leads us to the details of this person's account.

5.Vulnerability Name: Insecure Direct object Reference

CWE: 639

OWASP Category: A01: Broken Access Control

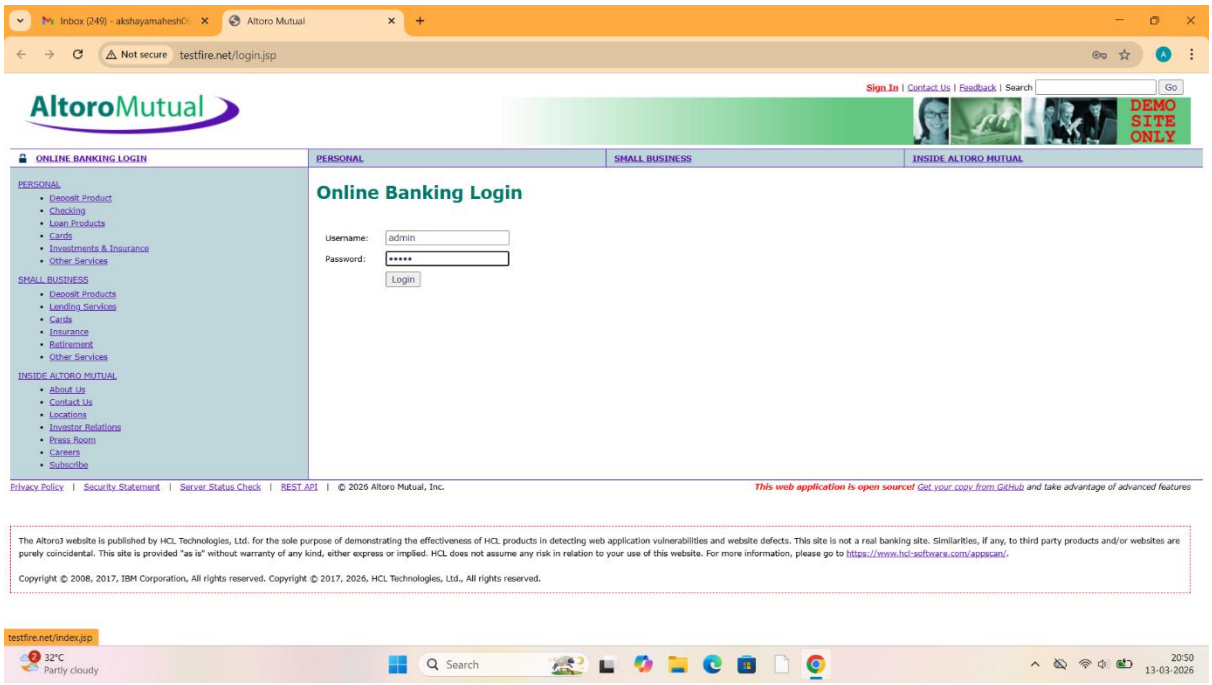
Description:

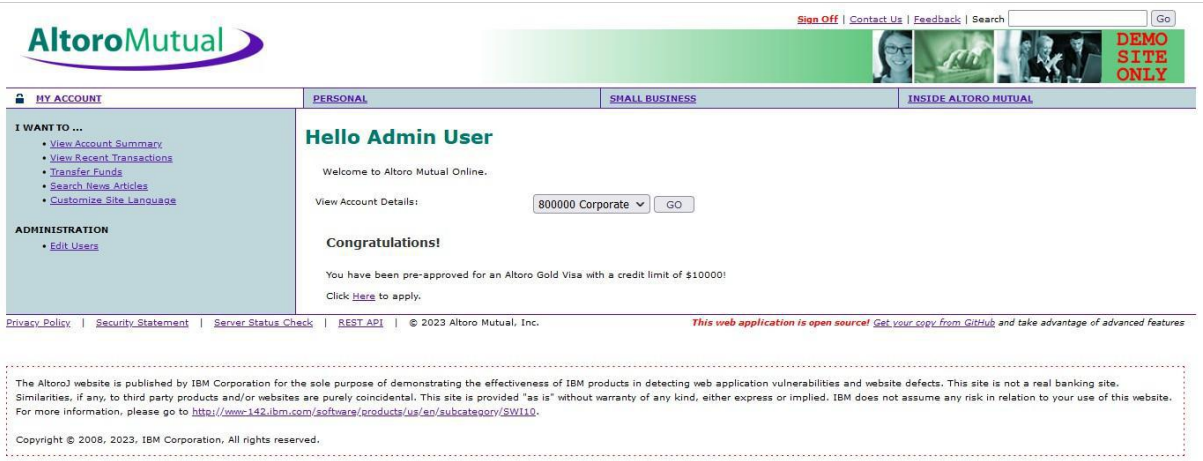
Insecure Direct Object Reference (IDOR) is a vulnerability that arises when attackers can access or modify objects by manipulating identifiers used in a web application's URLs or parameters. It occurs due to missing access control checks, which fail to verify whether a user should be allowed to access specific data.

Business Impact:

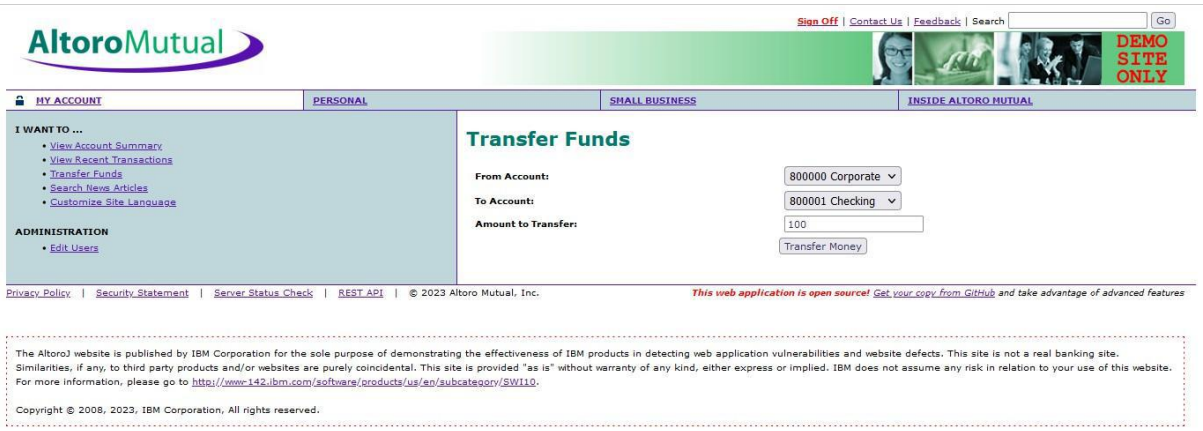
To effectively mitigate the business impact of CWE-639, organizations must make it a top priority to fortify their access control and authorization mechanisms. This entails the implementation of robust security measures, the regular conduct of comprehensive security assessments, and the deployment of intrusion detection systems to promptly identify and counter unauthorized access attempts. These measures stand as absolutely critical in the defense of sensitive data, the overall security of systems, and the preservation of the organization's esteemed reputation.

Sign in to the site of domain:





Open Transfer Money and click Intercept:



Open burp and notice the change:

```
Pretty   Raw   Hex
1  POST /bank/doTransfer HTTP/1.1
2  Host: testfire.net
3  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/119.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Content-Type: application/x-www-form-urlencoded
8  Content-Length: 78
9  Origin: http://testfire.net
10 Connection: close
11 Referer: http://testfire.net/bank/transfer.jsp
12 Cookie: JSESSIONID=17BCE438BBB750A14D269616C7623BB0; AltoroAccounts=
   "ODAwMDAwfkNvcnBvcmF0ZX4xLjYzNTczNTM1NjM10DcONEUy0Xw4MDAwMDF+Q2hlY2tpbmd+LTEuNjM1NzM1MzU2MzU4NzQORTI5fA=="
13 Upgrade-Insecure-Requests: 1
14
15 fromAccount=800000&toAccount=800001&transferAmount=100&transfer=Transfer+Money
```

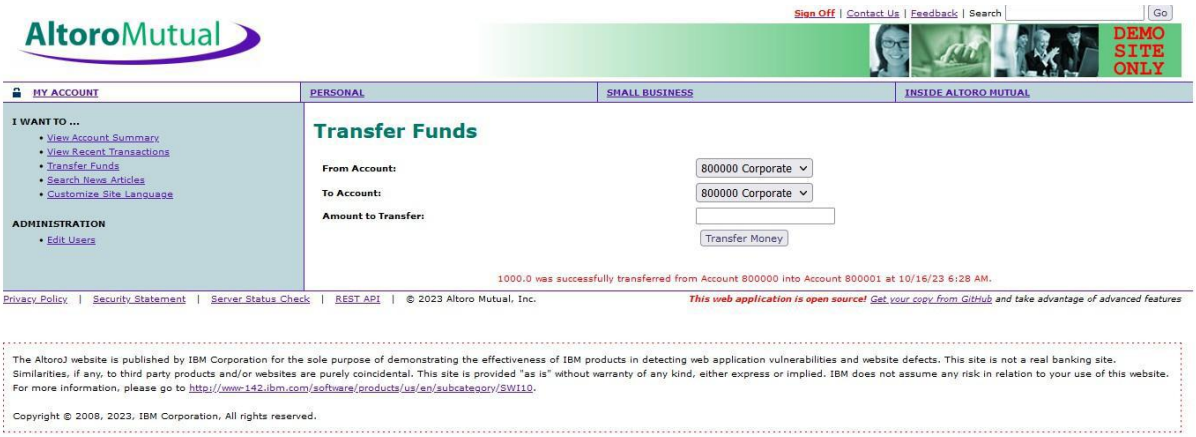
15th line:

```
fromAccount=800000&toAccount=800001&transferAmount=100&transfer=Transfer+Money
```

```
fromAccount=800000&toAccount=800001&transferAmount=1000&transfer=Transfer+Money
```

```
Pretty   Raw   Hex
1  GET /v1/tiles HTTP/1.1
2  Host: contile.services.mozilla.com
3  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/119.0
4  Accept: */*
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Te: trailers
8  Connection: close
9
10
```

We can notice the msg that shows the transfer of 1000:



Off the Intercept and open “View Recent Transactions”:

I WANT TO ...

- View Account Summary
- View Recent Transactions
- Transfer Funds
- Search News Articles
- Customize Site Language

ADMINISTRATION

- Edit Users

Recent Transactions

After yyyy-mm-dd Before yyyy-mm-dd Submit

Transaction ID	Transaction Time	Account ID	Action	Amount
9016	2023-10-16 06:28	800001	Deposit	$1000.00
9015	2023-10-16 06:28	800000	Withdrawal	-$1000.00
8920	2023-10-16 04:46	800001	Deposit	$23345.00
8919	2023-10-16 04:46	800000	Withdrawal	-$23345.00
8916	2023-10-16 04:44	800001	Deposit	$23345.00
8915	2023-10-16 04:44	800000	Withdrawal	-$23345.00
8894	2023-10-16 04:40	800001	Deposit	$23345.00
8893	2023-10-16 04:40	800000	Withdrawal	-$23345.00
8892	2023-10-16 04:40	800001	Deposit	$23345.00
8891	2023-10-16 04:40	800000	Withdrawal	-$23345.00
7565	2023-10-16 04:17	800000	Withdrawal	-$10.00
7228	2023-10-16 04:14	800000	Deposit	$1000.00
6802	2023-10-16 04:09	800001	Deposit	$100000000000000.00
6801	2023-10-16 04:09	800000	Withdrawal	-$100000000000000.00
5388	2023-10-16 03:58	800000	Deposit	$10000000000.00
4628	2023-10-16 03:36	800001	Deposit	$20190.00
4627	2023-10-16 03:36	800000	Withdrawal	-$20190.00
4626	2023-10-16 03:35	800001	Deposit	$20190.00
4625	2023-10-16 03:35	800000	Withdrawal	-$20190.00
4624	2023-10-16 03:35	800001	Deposit	$600.00
4623	2023-10-16 03:35	800000	Withdrawal	-$600.00
4616	2023-10-16 03:10	800000	Deposit	$87446.00
4615	2023-10-16 03:10	800001	Withdrawal	-$87446.00

6.Vulnerability name: Security Misconfiguration

CWE:732

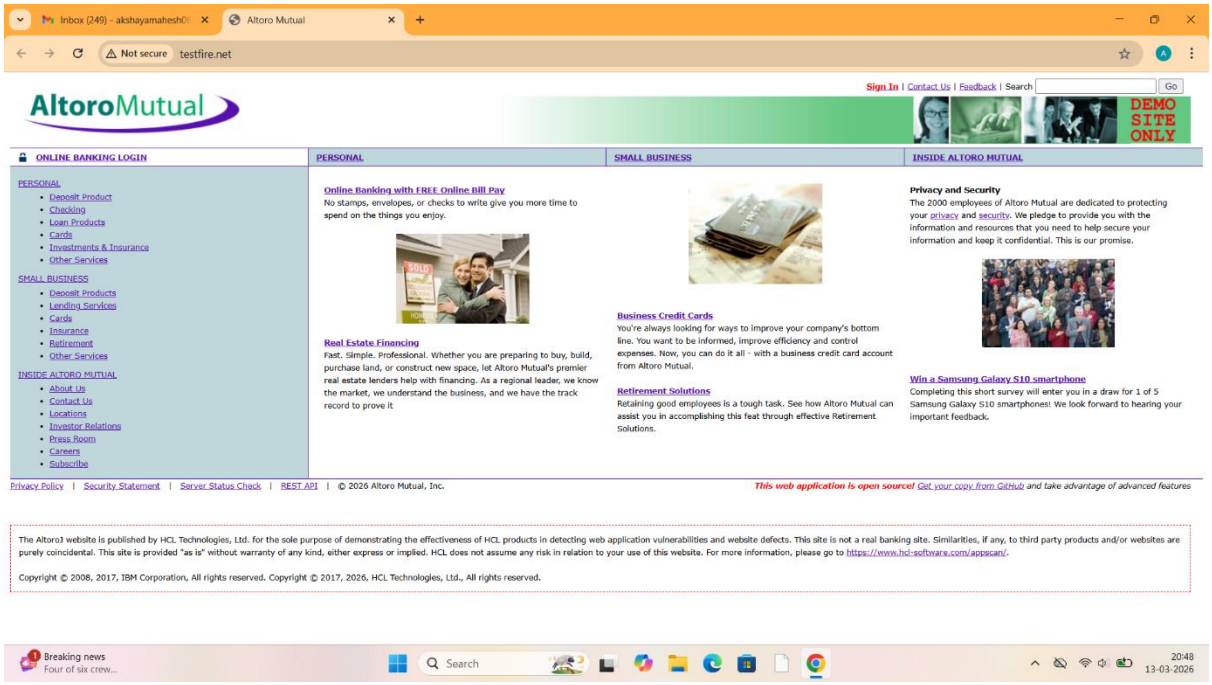
Description:

The product specifies permissions for a security-critical resource in a way that allows that resource to be read or modified by unintended actors.

Business Impact:

To effectively lessen the business consequences stemming from CWE-732, organizations must place a strong emphasis on bolstering their configuration management practices. This involves the establishment of secure default settings, the routine conduct of comprehensive security evaluations, and the deployment of intrusion detection systems to promptly detect and address misconfigurations and vulnerabilities. These actions are of paramount importance in safeguarding sensitive data, fortifying system security, and preserving the organization's esteemed reputation.

Access the URL:



Using the uniscan tool which is available in the kali:

```
┌──(kali㉿kali)-[~]
└─$ sudo uniscan -h
[sudo] password for kali:
#####################################
# Uniscan project                   #
# http://uniscan.sourceforge.net/   #
#####################################
V. 6.3

OPTIONS:
        -h      help
        -u      <url> example: https://www.example.com/
        -f      <file> list of url's
        -b      Uniscan go to background
        -q      Enable Directory checks
        -w      Enable File checks
        -e      Enable robots.txt and sitemap.xml check
        -d      Enable Dynamic checks
        -s      Enable Static checks
        -r      Enable Stress checks
        -i      <dork> Bing search
        -o      <dork> Google search
        -g      Web fingerprint
        -j      Server fingerprint

usage:
[1] perl ./uniscan.pl -u http://www.example.com/ -qweds
[2] perl ./uniscan.pl -f sites.txt -bqweds
[3] perl ./uniscan.pl -i uniscan
[4] perl ./uniscan.pl -i "ip:xxx.xxx.xxx.xxx"
[5] perl ./uniscan.pl -o "inurl:test"
[6] perl ./uniscan.pl -u https://www.example.com/ -r
```

We will use this tool to scan for hidden directories :

```
┌──(kali㉿kali)-[~]
└─$ sudo uniscan -u http://testfire.net/ -qweds
[sudo] password for kali:
#####################################
# Uniscan project                   #
# http://uniscan.sourceforge.net/   #
#####################################
V. 6.3


Scan date: 17-10-2023 7:18:22
 ==================================================================
 | Domain: http://testfire.net/
 | Server: Apache-Coyote/1.1
 | IP: 65.61.137.117
 ==================================================================
 |
 | Directory check:
 | [+] CODE: 200 URL: http://testfire.net/admin/fckeditor/admins/
 | [+] CODE: 200 URL: http://testfire.net/admin/
 | [+] CODE: 200 URL: http://testfire.net/admin/scripts/fckeditor/
 | [+] CODE: 200 URL: http://testfire.net/admin/FCKeditor/
 | [+] CODE: 200 URL: http://testfire.net/aux/
 | [+] CODE: 200 URL: http://testfire.net/bank/
 ==================================================================
 |
```

We found directories that can be accessed easily. Also all the files that are present in the website has also been found:

```
File check:
[+] CODE: 200 URL: http://testfire.net/admin/account.asp
[+] CODE: 200 URL: http://testfire.net/admin/account.html
[+] CODE: 200 URL: http://testfire.net/admin/admin.php
[+] CODE: 200 URL: http://testfire.net/admin/admin_phpinfo.php4
[+] CODE: 200 URL: http://testfire.net/admin/account.php
[+] CODE: 200 URL: http://testfire.net/admin/admin.shtml
[+] CODE: 200 URL: http://testfire.net/admin/aindex.htm
[+] CODE: 200 URL: http://testfire.net/admin/auth.php
[+] CODE: 200 URL: http://testfire.net/admin/cfg/configscreen.inc.php+
[+] CODE: 200 URL: http://testfire.net/admin/cfg/configsite.inc.php+
[+] CODE: 200 URL: http://testfire.net/admin/cfg/configsql.inc.php+
[+] CODE: 200 URL: http://testfire.net/admin/cfg/configtache.inc.php+
[+] CODE: 200 URL: http://testfire.net/admin/cms/htmltags.php
[+] CODE: 200 URL: http://testfire.net/admin/config.php
[+] CODE: 200 URL: http://testfire.net/admin/contextAdmin/contextAdmin.html
[+] CODE: 200 URL: http://testfire.net/admin/controlpanel.asp
[+] CODE: 200 URL: http://testfire.net/admin/controlpanel.php
[+] CODE: 200 URL: http://testfire.net/admin/controlpanel.html
[+] CODE: 200 URL: http://testfire.net/admin/cplogfile.log
[+] CODE: 200 URL: http://testfire.net/admin/cp.php
[+] CODE: 200 URL: http://testfire.net/admin/cp.html
[+] CODE: 200 URL: http://testfire.net/admin/database/wwForum.mdb
[+] CODE: 200 URL: http://testfire.net/admin/credit_card_info.php
[+] CODE: 200 URL: http://testfire.net/admin/datasource.asp
[+] CODE: 200 URL: http://testfire.net/admin/db.php
[+] CODE: 200 URL: http://testfire.net/admin/home.asp
[+] CODE: 200 URL: http://testfire.net/admin/home.php
[+] CODE: 200 URL: http://testfire.net/admin/index.asp
[+] CODE: 200 URL: http://testfire.net/admin/index.html
[+] CODE: 200 URL: http://testfire.net/admin/index.php
[+] CODE: 200 URL: http://testfire.net/admin/phpinfo.php
[+] CODE: 200 URL: http://testfire.net/admin/settings.inc.php+
[+] CODE: 200 URL: http://testfire.net/admin/system_footer.php
[+] CODE: 200 URL: http://testfire.net/admin/upload.php
[+] CODE: 200 URL: http://testfire.net/admin/templates/header.php
[+] CODE: 200 URL: http://testfire.net/admin/wg_user-info.ml
[+] CODE: 200 URL: http://testfire.net/admin/script.php
[+] CODE: 200 URL: http://testfire.net/login.jsp
```

Let us go through one webpage to see the vulnerability :

We can see the sensitive data of job applications which can help an attacker.

7.Vulnerability Name: Cross-Site Request Forgery

CWS: 352

OWASP Category: A08: Cross-Site Request Forgery

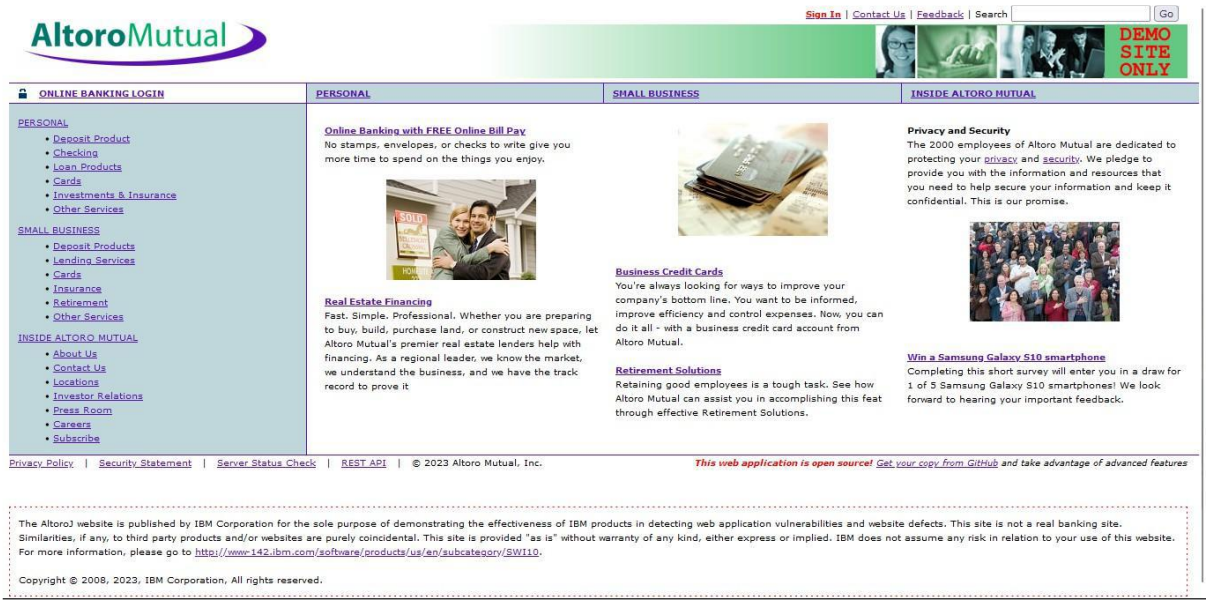
Description:

Cross-Site Request Forgery (CSRF) is an attack that forces an end user to execute unwanted actions on a web application in which they're currently authenticated. With a little help of social engineering (such as sending a link via email or chat), an attacker may trick the users of a web application into executing actions of the attacker's choosing. If the victim is a normal user, a successful CSRF attack can force the user to perform state changing requests like transferring funds, changing their email address, and so forth. If the victim is an administrative account, CSRF can compromise the entire web application.

Business Impact:

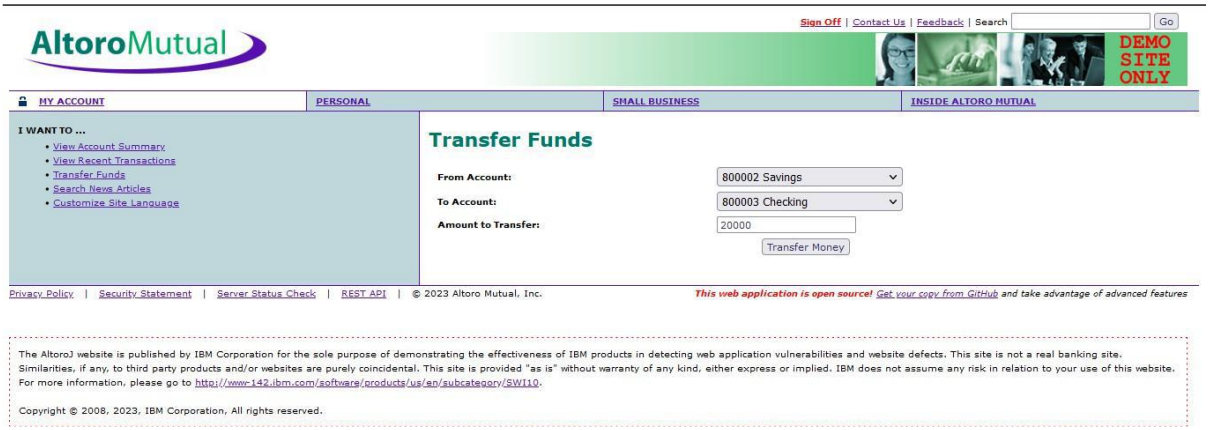
To effectively reduce the business repercussions of CWE-352, organizations must prioritize secure coding practices, including the incorporation of anti-CSRF tokens, while also conducting routine security assessments and testing to detect and rectify CSRF vulnerabilities. Furthermore, educating users on secure browsing practices is pivotal in preventing CSRF attacks. These actions are of paramount importance in the protection of data and in upholding the trust of customers and partners.

Accessing the URL:

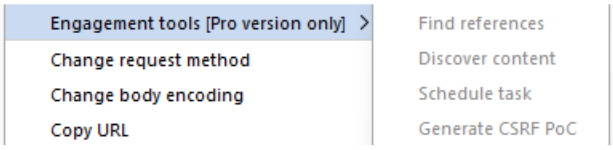


Login to it using default credentials, and you will find two accounts:

Capture the request in any Intercepting proxy (Burp Suite, Charles Proxy etc.) while transferring the amount from one account to another as shown in the below picture:



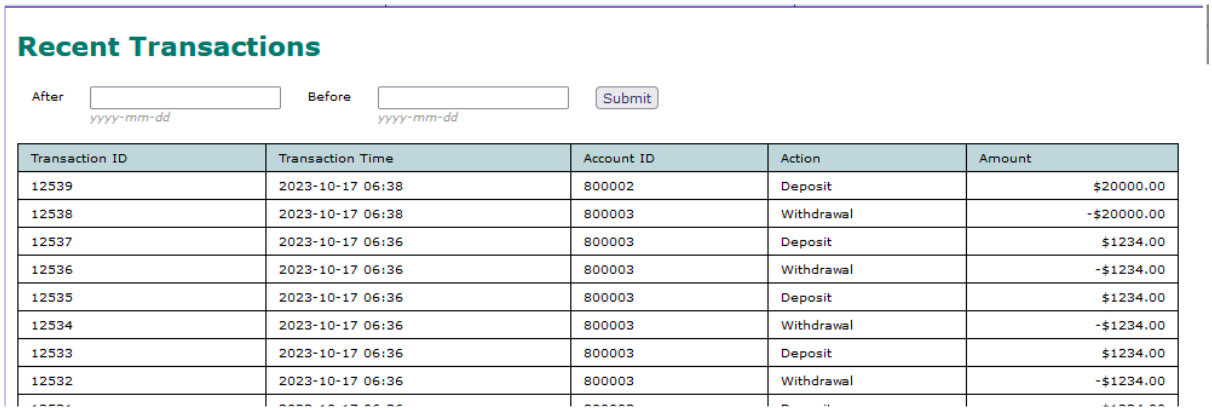
Generate CSRF POC using Burp Suite Engagement tools :



As the victim is already logged-in to the site an attacker can send the crafted malicious form/URL to the user to his email/text/by any other means and trick the victim using his social engineering skills/techniques :

```
Limited Time Offer Get a $100 Gift Card Now!!  (Evil :P)
```

User thinks, he could use a free $100 gift card! What could go wrong? You open the email and click the link/open the attachment. Once the user opens the form a transaction is automatically performed from his account without his knowledge as the user is already logged in.



Recent Transactions

After yyyy-mm-dd Before yyyy-mm-dd Submit

Transaction ID	Transaction Time	Account ID	Action	Amount
12539	2023-10-17 06:38	800002	Deposit	$20000.00
12538	2023-10-17 06:38	800003	Withdrawal	-$20000.00
12537	2023-10-17 06:36	800003	Deposit	$1234.00
12536	2023-10-17 06:36	800003	Withdrawal	-$1234.00
12535	2023-10-17 06:36	800003	Deposit	$1234.00
12534	2023-10-17 06:36	800003	Withdrawal	-$1234.00
12533	2023-10-17 06:36	800003	Deposit	$1234.00
12532	2023-10-17 06:36	800003	Withdrawal	-$1234.00

8.Vulnerability name: Cleartext Transmission of Sensitive Information

CWE: 319

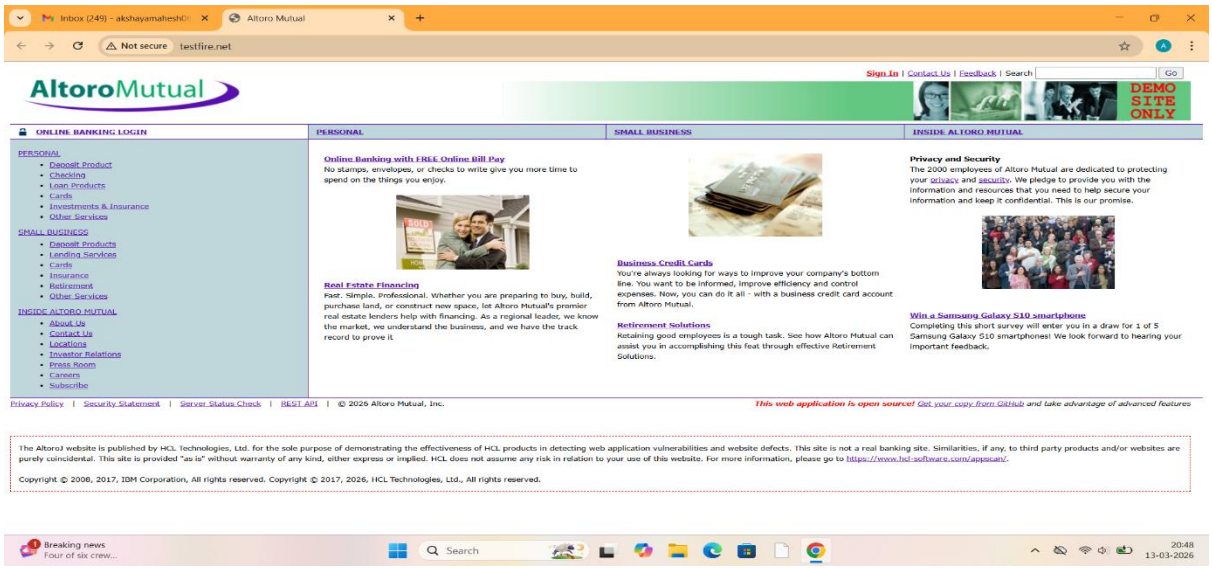
Description:

The product transmits sensitive or security-critical data in cleartext in a communication channel that can be sniffed by unauthorized actors. Many communication channels can be "sniffed" (monitored) by adversaries during data transmission. For example, in networking, packets can traverse many intermediary nodes from the source to the destination, whether across the internet, an internal network, the cloud, etc. Some actors might have privileged access to a network interface or any link along the channel, such as a router, but they might not be authorized to collect the underlying data. As a result, network traffic could be sniffed by adversaries, spilling security-critical data.

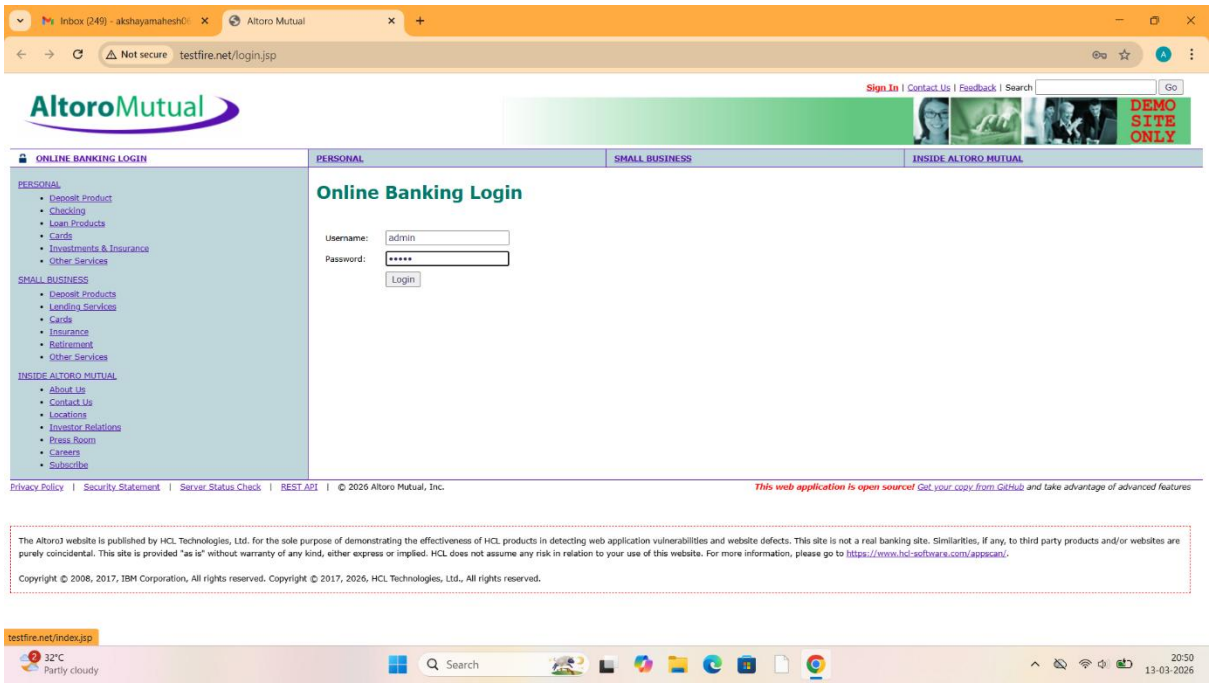
Business Impact:

To effectively reduce the business consequences associated with CWE-319, organizations must prioritize the adoption of secure data transmission practices. This includes the utilization of encryption and robust, secure protocols. The routine conduct of security assessments and testing is pivotal in pinpointing and remedying vulnerabilities linked to data transmission. Furthermore, the education of users on secure data handling practices plays a vital role in proactively preventing data exposure incidents. These actions are of paramount importance in the protection of sensitive data and in preserving the trust of both customers and partners.

Access the URL:



Sign in:



Request has been received in the burp suite with the username and password as well in clear text:

Pretty Raw Hex

```
POST /doLogin HTTP/1.1
Host: testfire.net
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/117.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Content-Type: application/x-www-form-urlencoded
Content-Length: 37
Origin: http://testfire.net
Connection: close
Referer: http://testfire.net/login.jsp
Cookie: JSESSIONID=B6ADB36ACD5C83083787343A1F97F853; AltoroAccounts="ODAwMDAwfkNvcnBvcmFOZX45LjQ3OTAlMTE2MUU3fDgwMDAwMX5DaGVjaZluZ34tNC4yMjc0NjkONTZFN3w="
Upgrade-Insecure-Requests: 1

uid=admin&passw=admin&btnSubmit=Login
```

9.Vulnerability name: Click Jacking

CWE: 1021

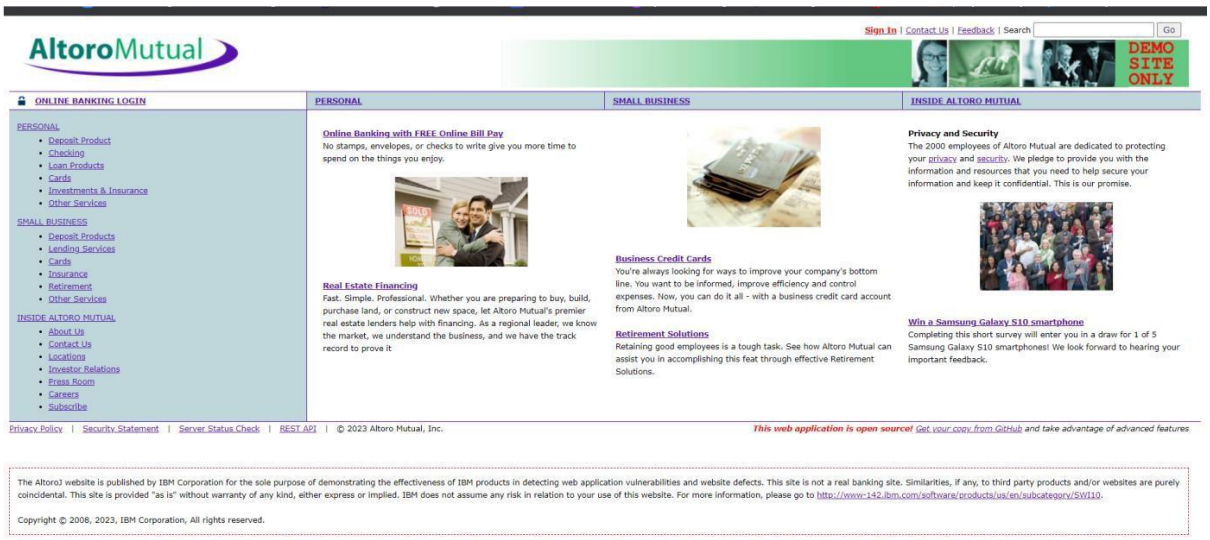
Description:

The web application does not restrict or incorrectly restricts frame objects or UI layers that belong to another application or domain, which can lead to user confusion about which interface the user is interacting with**.**

Business Impact:

To effectively reduce the business consequences resulting from CWE-1021, organizations must prioritize the implementation of protective measures, such as frame-busting code. Simultaneously, educating users on safe browsing practices plays a crucial role in preventing clickjacking incidents. Additionally, routine security assessments and testing are pivotal for identifying and mitigating vulnerabilities associated with clickjacking. These actions are of paramount importance in upholding user trust, ensuring data protection, and safeguarding the organization's reputation**.**

Access the URL:

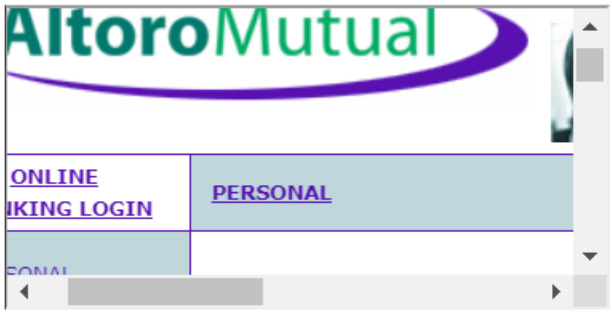


Write the HTML code:

```html
<html>
<body>
<title>Click jacking vulnerability</title>
<h2>This website is vulnerable to clickjacking</h2>
<iframe  src="http://testfire.net/"></iframe>
</body>
</html>
```

Execution of the Code:

This website is vulnerable to clickjacking



From this image we can see that the vulnerability has been found.

10.Vulnerability name: web server allows password auto-completion

CWE: 310

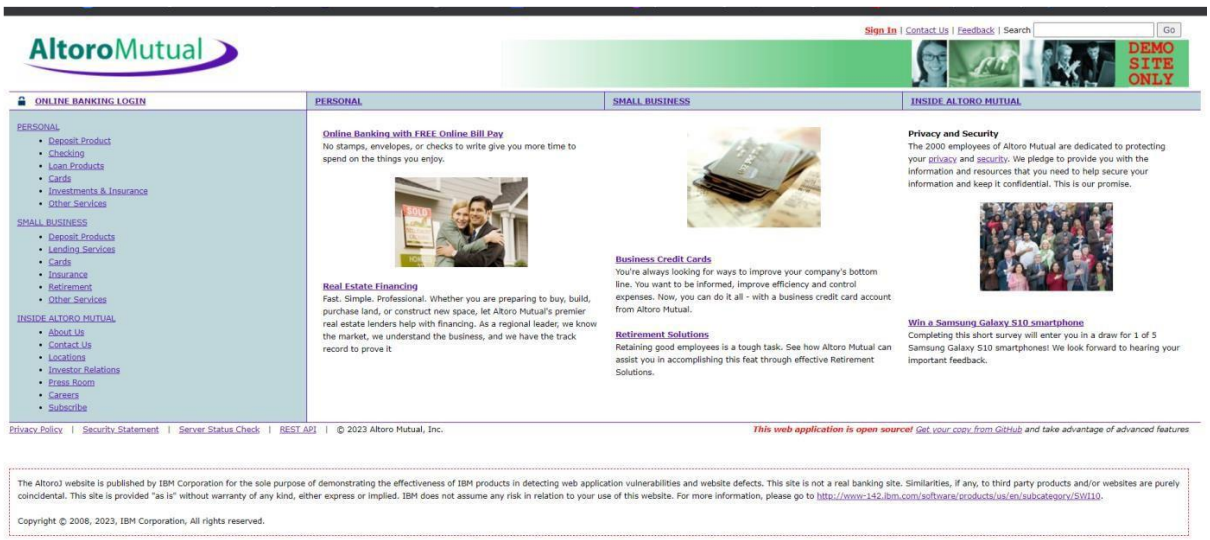
Description:

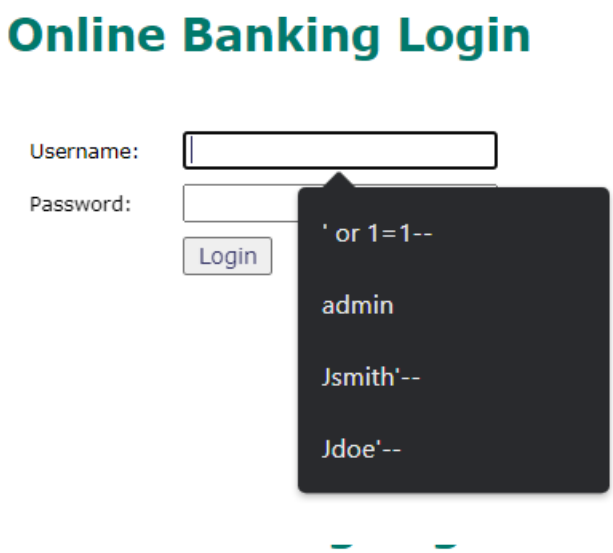
Weaknesses in this category are related to the design and implementation of data confidentiality and integrity. Frequently these deal with the use of encoding techniques, encryption libraries, and hashing algorithms. The weaknesses in this category could lead to a degradation of the quality data if they are not addressed.

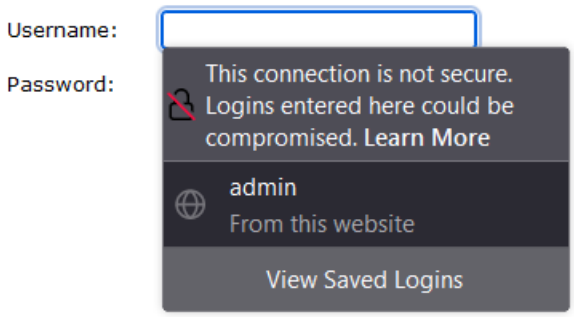
Business impact:

To effectively lessen the business consequences associated with CWE-310, organizations must prioritize the adoption of secure cryptographic practices. This entails ensuring proper password storage and robust encryption key management. Simultaneously, conducting routine security assessments and testing is pivotal in detecting and mitigating vulnerabilities related to cryptographic issues. Striving for compliance with pertinent data protection regulations and industry standards is equally essential. These actions are of paramount importance in the protection of sensitive data, the upholding of user trust, and the preservation of the organization's esteemed reputation.

Access the URL:







From this image we can see the usernames and the passwords getting auto filled. This is a potential vulnerability as this can be a doorway for attackers.

Stage-2

Overview :

Nessus is a widely used vulnerability assessment tool developed by Tenable that helps organizations identify security weaknesses in their networks, systems, and applications. It is designed to scan computers, servers, databases, web applications, and network devices to detect vulnerabilities that could be exploited by attackers. Nessus uses a large and frequently updated database of vulnerability plugins to check for known security issues such as missing patches, outdated software versions, insecure configurations, open ports, weak encryption protocols, and exposed services. The tool performs automated scans on specified targets and analyzes the system for potential risks. After the scan is completed, Nessus generates detailed reports that categorize vulnerabilities based on severity levels such as informational, low, medium, high, and critical. These reports also include explanations of the vulnerabilities, their potential impact, and recommendations for remediation or mitigation.

In addition, Nessus supports both authenticated and unauthenticated scanning. Authenticated scans allow the tool to log into the system using credentials and perform deeper analysis, which results in more accurate vulnerability detection. Nessus also helps organizations comply with various security standards and regulations by identifying configuration weaknesses and policy violations. Security analysts, penetration testers, and IT administrators commonly use Nessus as part of their vulnerability management process to regularly monitor systems, prioritize security risks, and fix vulnerabilities before they can be exploited. Because of its reliability, comprehensive vulnerability coverage, and user-friendly interface, Nessus has become one of the most trusted and widely used vulnerability scanning tools in the field of cybersecurity.

Target website: testfire.net

Target ip address: 54.83.41.36

List of vulnerability :

S.No	Vulnerability name	Severity	plugins
1	Cross-site scripting (stored)	Low	Web application Scanner
2	Broken Access Control	Medium	SSH weak HMAC Algorithms Supported
3	SQL Injection	Medium	N/A
4	Broken Authentication	Low	HSTS Missing from Server
5	Insecure Direct Object Reference	Info	Service Detection
6	Security Misconfiguration	Low	SSL cipher Block Chaining cipher suite
7	Cross-Site Request Forgery	Low	HTTP Form- Based Authentication Detected
8	Cleartext Transmission of Sensitive Information	Medium	Cleartext Transmission of Sensitive Information over HTTP
9	ClickJacking	Info	HTML Rendering Layer Vulnerability
10	webserver allows password auto- completion	Low	SSL/TLS Diffie-Hellman Modulus<=1024 bits(Logjam)

1.VULNERABILITY NAME: Cross site scripting (stored)

SEVERITY : Low

PLUGIN: web application scanner

PORT:N/A

DESCRIPTION:

It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

SOLUTION:

- When interacting with databases, use parameterized statements or prepared statements to Avoid SQL injection, which can be a vector for XSS.

  Sanitize and validate all user inputs and ensure that any data displayed on the web page properly encoded.
- Use output encoding libraries or functions to prevent script injection.

BUSINESS IMPACT:

The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area that is displayed to either many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data .

2. VULNERABILITY NAME: Broken access control

SEVERITY: medium

PLUGIN: SSH weak HMAC Algorithms supported

PORT: N/A

DESCRIPTION:

When access control checks are not applied consistently - or not at all users are able to access data or perform actions that they should not be allowed to perform. This can lead to a wide range of problems, including information exposures, denial of service, and arbitrary code execution.

SOLUTION:

- Ensure that the application has a robust authentication and authorization mechanism in place.
- Users should be authenticated before any sensitive actions are performed, and authorization checks should be conducted to verify that users have the necessary permissions for the requested operation.
- Ensure that session management is secure. Use strong session IDs, enforce session timeouts, and regenerate session tokens after login. Implement session fixation protection to prevent session hijacking.

BUSINESS IMPACT:

wide-ranging business impact, including financial losses, reputation damage legal consequences, operational disruption, and loss of customer trust. Addressing this vulnerability is crucial to protect the organization and its stakeholders from these potential negative effects.

3.VULNERABILITY NAME: SQL injection

SEVERITY : medium

PLUGIN: N/A (typically would be a security testing tool)

PORT: port 80/443

DESCRIPTION:

The product constructs all or part of an SQL command using externally influenced input from an upstream component, but it does not neutralize or incorrectly neutralizes special elements that could modify the intended SQL command when it is sent to a downstream component.

SOLUTION:

- If parameterized statements are not feasible, use proper input validation and escaping mechanisms to sanitize user inputs before they are used in SQL queries. This helps prevent malicious code injection.
- Utilize stored procedures for database operations. This can help separate SQL code from application code and reduce the risk of SQL injection.
- Implement strict input validation for all user inputs, ensuring that data adheres to the expected format and structure.

BUSINESS IMAPCT:

In summary, it is crucial to underscore that CWE-89, known as SQL Injection,can exert a profound and diverse business impact. This encompasses critical facets such data breaches, financial setbacks, harm to reputation, legal ramifications, and operational turmoil.

4. VULNERABILITY NAME: Broken authentication

SEVERITY: Low

PLUGIN: HSTS Missing from HTTPS Server (plugin id:142960)

PORT: port 443

DESCRIPTION:

The system's authorization functionality does not prevent one user from gaining access to another user's data or record by modifying the key value identifying the data.

SOLUTION:

•Implement strong authentication mechanisms, such as multi-factor authentication (MFA or two-factor authentication (2FA), to ensure that only authorized users can access the system.

- Implement proper authorization controls to ensure that users have the necessary permissions to access specific functions or data. Use role-based access control (RBAC) to manage user privileges effectively.
- Deploy intrusion detection and prevention systems (IDS/IPS) to monitor and block suspicious activities related to authentication and authorization.

BUSINESS IMPACT:

To effectively mitigate the business impact of CWE-285, it is imperative that organizations place a strong emphasis on bolstering their authentication and session management practices. This should encompass the adoption of multi-factor authentication, the secure storage of credentials, meticulous session handling, and a commitment to conducting routine security assessments and testing.

5.VULNERABILITY NAME: Insecure Direct object Reference

SEVERITY: Info

PLUGIN: Service Detection (plugin id:22964)

PORT: port 80/443

DESCRIPTION:

Insecure Direct Object Reference (IDOR) is a vulnerability that arises whenattackers can access or modify objects by manipulating identifiers used in a web application'sURLs or parameters. It occurs due to missing access control checks, which fail to verify whether a user should be allowed to access specific data.

SOLUTION:

- Utilize parameterized queries or prepared statements when interacting with databases. This prevents user-controlled input from being executed as SQL code.
- Implement a Content Security Policy to mitigate the risk of cross-site scripting (XSS) attacks, which can be used to execute unauthorized queries.
- Implement account lockout mechanisms that temporarily suspend accounts after a specified number of unauthorized access attempts to prevent brute force attacks.

BUSINESS IMPACT:

To effectively mitigate the business impact of CWE-639, organizations must make it a top priority to fortify their access control and authorization mechanisms. ThiSentails the implementation of robust security measures, the regular conduct of comprehensive security assessments, and the deployment of intrusion detection systems to promptly identify and counter unauthorized access attempts. These measures stand as critical in the defence of sensitive data, the overall security of systems, and the preservation of the organization's esteemed reputation.

6.VULNERABILITY NAME: Security Misconfiguration

SEVERITY: low

PLUGIN:SSL cipher Block chaining cipher suites

PORT: port 443

DESCRIPTION:

The product specifies permissions for a security-critical resource in a way that allows that resource to be read or modified by unintended actors.

SOLUTION:

- Implement a change management process to document and review all changes to system configurations. This helps prevent unauthorized or unintended alterations.
- Follow security guidelines provided by software and hardware vendors to ensure secure configurations.
- Regularly perform compliance checks against industry standards and regulations to ensure that configurations remain in compliance with security requirements.

BUSINESS IMPACT:

To effectively lessen the business consequences stemming from CWE-732, organizations must place a strong emphasis on bolstering their configuration management practices. This involves the establishment of secure default settings, the routine conduct of comprehensive security evaluations, and the deployment of intrusion detection systems to promptly detect and address misconfigurations and vulnerabilities. These actions are of paramount importance in safeguarding sensitive data, fortifying system security, and preserving the organization's esteemed reputation.

7.VULNERABILITY NAME: Cross-Site Request Forgery

SEVERITY: Low

PLUGIN:HTTP Form-Based Authentication Detected

PORT: port 8080/443

DESCRIPTION:

Cross-Site Request Forgery (CSRF) is an attack that forces an end user to execute unwanted actions on a web application in which they're currently authenticated. With a little help of social engineering (such as sending a link via email or chat), an attacker may trick the users of a web application into executing actions of the attacker's choosing. If the victim is a normal user, a successful CSRF attack can force the user to perform state changing requests like transferring funds, changing their email address, and so forth. If the victim is an administrative account, CSRF can compromise the entire web application.

SOLUTION:

- Implement Synchronizer Token Patterns to include unique, randomly generated tokens in each HTTP request and validate these tokens on the server side to verify the authenticity of the request.
- Utilize the Same Site attribute in cookies to restrict their usage to same-site requests only, preventing them from being sent along with cross-site requests and effectively mitigating CSRF attacks.
- Utilize framework-specific protections and security features to prevent CSRF attacks. Many modern web frameworks have built-in mechanisms and libraries for handling CSRF vulnerabilities effectively.
- Implement a Content Security Policy to restrict the sources from which various types of content can be loaded. By specifying the trusted sources of content, you can minimize the risk of malicious code execution and reduce the likelihood of successful CSRF attacks.

BUSINESS IMPACT:

To effectively reduce the business repercussions of CWE-352, organization must prioritize secure coding practices, including the incorporation of anti-CSRF tokens, while also conducting routine security assessments and testing to detect and rectify CSRF vulnerabilities. Furthermore, educating users on secure browsing practices is pivotal in preventing CSRF attacks. These actions are of paramount importance in the protection of data and in upholding the trust of customers and partners.

8. VULNERABILITY NAME: Cleartext Transmission of Sensitive Information

SEVERITY: Medium

PLUGIN: Cleartext Transmission of Sensitive Information over HTTP

PORT: port 443

DESCRIPTION:

The product transmits sensitive or security-critical data in cleartext in a communication channel that can be sniffed by unauthorized actors. Many communication channels can be "sniffed" (monitored) by adversaries during data transmission. For example, in networking, packets can traverse many intermediary nodes from the source to the destination, whether across the internet, an internal network, the cloud, etc. Some actors might have privileged access to a network interface or any link along the channel, such as a router, but they might not be authorized to collect the underlying data. As a result, network traffic could be sniffed by adversaries, spilling securitycritical data.

SOLUTION:

- Educate users and employees on secure data handling practices, such as recognizing secure websites (look for "https") and avoiding insecure Wi-Fi networks.
- Employ data masking or redaction techniques to replace sensitive data with placeholders or cryptographic representations during transmission.
- Implement encryption for data in transit. Use secure encryption protocols such as TLS/SSL to protect sensitive information during transmission.
- Use secure communication protocols for transmitting data, such as HTTPS for web applications, and ensure that the selected protocols are kept up to date.

BUSINESS IMPACT:

To effectively reduce the business consequences associated with CWE-319,organizations must prioritize the adoption of secure data transmission practices. This includes the utilization of encryption and robust, secure protocols. The routine conduct of security assessments and testing is pivotal in pinpointing and remedying vulnerabilities linked to data transmission. Furthermore, the education of users on secure data handling practices plays a vital role in proactively preventing data exposure incidents. These actions are of paramount importance in the protection of sensitive data and in preserving the trust of both customers and partners.

9.VULNERABILITY NAME: Clickjacking

SEVERITY: Info

PLUGIN: HTML Rendering Layer Vulnerability

PORT: port 80/443

DESCRIPTION:

The web application does not restrict or incorrectly restricts frame objects or UI layers that belong to another application or domain, which can lead to user confusion about which interface the user is interacting with.

SOLUTION:

- Set the X-Frame-Options HTTP response header to deny or same-origin to control how your site can be framed. This is supported by most modern browsers.
- Utilize frame-busting JavaScript code in web applications to prevent the embedding of your site within malicious frame's. This code can disrupt clickjacking attempts.
- Implement a Content Security Policy to restrict which domains can embed your site in iframes. This can help prevent unauthorized framing.
- Implement additional security controls to prevent UI redress attacks, such as clickjacking, within your web application.

BUSINESS IMPACT:

To effectively reduce the business consequences resulting from CWE-1021,organizations must prioritize the implementation of protective measures, such as frame-busting code. Simultaneously, educating users on safe browsing practices plays a crucial role in preventing clickjacking incidents. Additionally, routine security assessments and testing are pivotal for identifying and mitigating vulnerabilities associated with clickjacking. These actions are of paramount importance in upholding user trust, ensuring data protection, and safeguarding the organization's reputation.

10.VULNERABILITY NAME: web server allows password auto-completion

SEVERITY: Low

PLUGIN: SSL/TLS Diffie-Hellman Modulus <= 1024 Bits (Logjam)

PORT: port 443

DESCRIPTION:

Weaknesses in this category are related to the design and implementation of data confidentiality and integrity. Frequently these deal with the use of encoding techniques, encryption libraries, and hashing algorithms. The weaknesses in this category could lead to a degradation of the quality data if they are not addressed.

SOLUTION:

- Implement secure password storage mechanisms, such as using strong and salted cryptographic hashing algorithms like crypt or script.
- Implement robust encryption key management practices, including secure key storage, key rotation, and access controls.
- Utilize well-established and proven cryptographic libraries and algorithms for encryption and decryption.

BUSINESS IMPACT:

To effectively lessen the business consequences associated with CWE-310, organizations must prioritize the adoption of secure cryptographic practices. This entails ensuring proper password storage and robust encryption key management. Simultaneously, conducting routine security assessments and testing is pivotal in detecting and mitigating vulnerabilities related to cryptographic issues. Striving for compliance with pertinent data protection regulations and industry standards is equally essential. These actions are of paramount importance in the protection of sensitive data, the upholding of user trust, and the preservation of the organization's esteemed reputation.

S.No	Vulnerability Name	Severity	Plugins
1	Cross-site scripting (DOM based XSS)	Medium	Multiple XSS
2	jQuery 1.3<3.5.0	Medium	CGI Abuse: XSS
3	information exposure (web server HTTP header internal IP disclosure)	Low	Webserver HTTP Header Internal IP Disclosure
4	Information exposure (SSL (Multiple Issues))	Info	SSL Certificate Information
5	Information exposure (HTTP (Multiple Issues))	Info	HTTP Information
6	OS identification failed	Info	TCP/IP Remote Probing

TARGET WEBSITE: vit.ac.in

TARGET IP ADDRESS:122.184.65.22

Report:

1. VULNERABILITY NAME: Cross-site scripting (DOM based XSS)

SEVERITY: Medium

PLUGIN: Multiple XSS

PORT: port 443

DESCRIPTION:

It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser. DOMXSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document, and this document can be either displayed in the browser window or as the HTML Source.

SOLUTION:

- When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
- Sanitize and validate all user inputs and ensure that any data displayed on the web page is properly encoded. Use output encoding libraries or functions to prevent script injection.
- Avoid using DOM manipulation methods that can introduce vulnerabilities. Be cautious with inner HTML, document. Write, and other methods that can execute scripts.
- Consider using well-established JavaScript frameworks and libraries that include built-in security features to prevent DOM XSS.

BUSINESS IMPACT:

The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

2. VULNERABILITY NAME: jQuery 1.2 < 3.5.0 Multiple XSS

SEVERITY: Medium

PLUGIN: CGI Abuse: XSS

PORT: port 443

DESCRIPTION:

It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser. Stored XSS (Persistent XSS): In this type of attack, the malicious script is permanently stored on a server or in a database, often in user-generated content like comments or forum posts. When other users view the affected content, the script executes in their browsers, potentially compromising their data or sessions. Reflected XSS: Reflected XSS occurs when the malicious script is embedded in a URL and immediately executed when a victim clicks on a manipulated link. The script is not stored on a server but is reflected off a web application, making it a one-time attack. DOM-based XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document, and this document can be either displayed in the browser window or as the HTML Source.

SOLUTION:

- The most important recommendation is to update to the latest version of jQuery. Newer versions typically include security patches, bug fixes, and performance improvements.
- Stay vigilant about jQuery updates and security announcements. Subscribe to relevant security mailing lists or forums to receive notifications about new releases.
- A Content Security Policy (CSP) can help mitigate various types of client-side vulnerabilities, including Cross-Site Scripting (XSS). Consider implementing a CSP in your web application to restrict the sources from which scripts can be loaded.
- If updating jQuery is not immediately possible, review your code for potential security issues. Ensure that you are escaping or sanitizing user inputs and that you are using safe coding practices to prevent XSS and other security vulnerabilities.

BUSINESS IMPACT:

The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

3. VULNERABILITY NAME: Information Exposure (Web Server HTTP Header Internal IP Disclosure)

SEVERITY: Low

PLUGIN: Web Server HTTP Header Internal IP Disclosure

PORT: port 80

DESCRIPTION:

The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

SOLUTION:

- Carefully review your web server's configuration files (e.g., Apache's httpd.conf, Nginx's nginx.conf) to identify anywhere internal IP addresses are inadvertently included in HTTP response headers.
- Remove or replace any internal IP addresses in the server configuration files with appropriate placeholders or public IP addresses.
- If you use proxy servers or load balancers, ensure that they are configured to hide internal IP addresses and only expose public IP addresses instances in HTTP headers.
- Review and configure HTTP response headers (e.g., Server, X-Powered-By) to limit the exposure of server and technology details, reducing the potential for disclosing internal IP addresses.

BUSINESS IMPACT:

To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry standards is imperative. These actions play a pivotal role in safeguarding sensitive data, upholding user trust, and safeguarding the organization's esteemed reputation.

4. VULNERABILITY NAME: Information Exposure (SSL (Multiple Issues))

SEVERITY: Info

PLUGIN: SSL certificate Information

PORT: port 443

DESCRIPTION:

The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

SOLUTION:

- Configure your web server to use the latest and strongest encryption protocols, such as TLS 1.2 or higher. Disable outdated and insecure protocols like SSLv3.
- Implement Perfect Forward Secrecy to ensure that even if the private key is compromised, previously recorded encrypted traffic remains secure.
- Safeguard your private keys by storing them securely and ensuring they are not accessible to unauthorized parties.

BUSINESS IMPACT:

To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry standards is imperative. These actions play a pivotal role in safeguarding sensitive data, upholding user trust, and safeguarding the organization's esteemed reputation.

5. VULNERABILITY NAME: Information Exposure (HTTP (Multiple Issues))

SEVERITY: Info

PLUGIN: HTTP information

PORT: port 80 /443

DESCRIPTION:

The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information

SOLUTION:

- Minimize the use of unnecessary redirects as they can increase page load times. Use 301 permanent) redirects for consistent content.
- Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
- Properly configure CORS to control which domains are allowed to make requests to your server.
- Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

BUSINESS IMPACT:

To effectively reduce the business repercussions of HTTP issues, organizations must place a strong emphasis on ongoing monitoring and enhancement of their web services. Consistent performance testing and security assessments, coupled with proactive maintenance, are indispensable to ensure a seamless and secure online presence. Furthermore, offering timely and helpful customer support and transparent communication when HTTP issues arise is instrumental in mitigating adverse effects on user trust and satisfaction.

6. VULNERABILITY NAME: OS Identification Failed

SEVERITY: Info

PLUGIN: TCP/IP remote probing

PORT: N/A

DESCRIPTION:

OS identification failed refers to the inability of a system or software to accurately recognize the operating system running on a device or network. This issue can result from various factors such as misconfigurations, network issues, or compatibility problems. The failure to identify the OS correctly can lead to security vulnerabilities, hindered software updates, and potential operational disruptions. Resolving this issue promptly is crucial to ensure accurate security measures and the smooth functioning of the system or network.

SOLUTION:

- Minimize the use of unnecessary redirects as they can increase page load times. Use 301 (permanent) redirects for consistent content.
- Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
- Properly configure CORS to control which domains are allowed to make requests to your server.
- Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

BUSINESS IMPACT:

It may lead to operational inefficiencies, potential security vulnerabilities, non-compliance with industry regulations, and a possible loss of customer trust. Such failures could result in downtime, compromised data security, regulatory penalties, and damage to the company's reputation, potentially leading to financial losses and decreased market competitiveness. Resolving this issue promptly is crucial to mitigate these risks and maintain the smooth functioning of the organization.

Stage-3

Title:

Design and apply context-aware access controls based on user identity, device posture, location, and application risk.

Context-aware access controls represent the operational pinnacle of a Zero Trust architecture, moving beyond static rules to create a dynamic and adaptive security perimeter around every application. In a modern enterprise, where users are mobile, devices are diverse, and threats are sophisticated, granting access based solely on a username and password is a dangerous oversimplification. Designing and applying context-aware controls means building a policy engine that consumes multiple, real-time signals about an access request and renders an intelligent decision: to allow, to block, to limit, or to challenge. This approach ensures that the security posture is not a flat, always-on gate but a responsive system that tightens or loosens based on the risk of the specific moment.

The design of such a system begins with a deep integration between the access broker, such as Zscaler Private Access (ZPA), and the organization's existing security and identity infrastructure. The first and most fundamental layer is user identity. This is not simply a username, but a rich set of attributes pulled from the corporate Identity Provider (IdP) like Azure AD or Okta. Integration via SAML ensures that when a user authenticates through the ZPA client, they are verified with multi-factor authentication (MFA) by the IdP. ZPA then ingests not just the user's name, but their group memberships (e.g., "Finance – Payroll Managers," "Engineering – Source Code Contributors"), roles, and department. This allows for the creation of granular policies that are intrinsically tied to the user's function within the organization.

For instance, a policy can be written in plain language: "Allow members of the 'Finance-Payroll' group to access the 'ADP Payroll' application." However, a valid identity on a compromised device is still a critical threat. This is where the second layer, device posture, becomes essential.

The design incorporates continuous checks on the security health of the endpoint attempting the connection. Through integrations with Endpoint Detection and Response (EDR) tools like CrowdStrike or SentinelOne, and Mobile Device Management (MDM) solutions like JAMF or Intune, ZPA can assess a device's real-time hygiene. The policy engine evaluates factors such as: Is the device's disk encryption (BitLocker/FileVault) enabled? Is the antivirus software active and its definition database up-to-date? Has the operating system been patched to the latest security baseline? Is the device rooted or jailbroken? Is there a high-risk score being reported by the EDR tool due to suspicious activity? If a device fails these checks—for example, a user's laptop has its firewall disabled—the context-aware policy can automatically trigger an action. It might block access entirely, or it might grant "remediation access," routing the user to a captive portal to download updates or contact IT, while preventing them from reaching any businesscritical applications. This ensures that a vulnerable endpoint cannot become a beachhead for an attacker to move laterally into the corporate environment.

The third contextual layer is location and network environment. The physical or virtual location from which a user connects provides a powerful signal for risk assessment. A finance director accessing the payroll system from their home office in the same city is a low-risk event. The same user attempting to connect from a country with a high prevalence of cybercrime, or from an anonymizing VPN service, should be treated with extreme suspicion. Context-aware policies can be designed to enforce this logic. For example, access to a source code repository containing intellectual property could be restricted to users connecting from a corporate office or a preapproved home IP address range. If a legitimate user is traveling, the policy might allow access but require a step-up authentication, such as a hardware token, or grant only read-only permissions. This layer can also consider the security of the network itself; a connection from a public, unencrypted Wi-Fi hotspot might trigger stricter controls than one from a known, secure corporate network.

The final and often most strategic layer is application risk. Not all applications are created equal, and their access policies should reflect their value and sensitivity. In the design phase, applications are categorized and tagged based on their risk profile. An internal wiki used for cafeteria menus would be classified as low-risk, whereas a customer database, a financial trading platform, or a source code repository would be classified as high or critical risk. This tagging allows policies to be layered and dynamic. A single user might have access to multiple applications, but the controls applied to each session will differ based on the application's risk. For the low-risk wiki, a valid user identity and a compliant device might be sufficient. However, for the high-risk financial platform, the policy engine might require all four layers to align: the user must be in a specific group, on a fully patched and managed device, connecting from a trusted location, and may even be subject to session recording or time-of-day restrictions. This ensures that security resources and scrutiny are applied proportionally to the actual risk, preventing "blanket" access and ensuring that the most valuable assets have the most robust defenses.

In practice, applying these controls within ZPA means using the policy builder to create conditional statements. A policy might read: "Allow access to 'Application A' if User is in 'Group X' AND Device Posture is 'Compliant' AND Location is 'Not Restricted' AND Application Risk is 'Low'." Alternatively, a more complex policy for a critical application might be: "Allow access to 'Application B' if User is in 'Group Y' AND Device Posture is 'Highly Secure' AND Location is 'Corporate Network' AND Time is 'Business Hours', otherwise, if Location is 'Remote', require Step-Up Authentication." By combining these signals, the organization creates a security posture that is finely tuned, highly resilient, and fundamentally adaptive to the context of every single access attempt, thereby upholding the core Zero Trust principle of "never trust, always verify."

Conclusion:

Stage 1: In conclusion, web application testing is a crucial process that ensures the security, functionality, and reliability of web-based software. It involves a comprehensive evaluation of various aspects, including user interface, functionality, performance, and security to identify and rectify issues before they impact end-users. This process not only enhances the quality of web applications but also bolsters their resilience against potential threats and vulnerabilities. In an age where online services and applications are integral to our daily lives, effective web application testing is essential to deliver a seamless and secure user experience.

Stage 2: The Nessus report for the specific website has highlighted key security insights. It identified vulnerabilities, categorized by severity, with a focus on critical and high-risk issues that need immediate attention. Recommendations for addressing these vulnerabilities include patching, reconfiguration, and other security measures, with a priority on actions that can have the most significant impact. Compliance with industry standards and best practices is considered, ensuring alignment with recognized security benchmarks like OWASP guidelines. The report underscores the need for continuous monitoring, regular updates, and ongoing security efforts to maintain a strong security posture. In summary, the Nessus report offers a clear view of the website's security state, provides guidance on mitigating vulnerabilities, and stresses the importance of sustained vigilance against evolving threats.

Stage 3: The implementation of Zscaler Private Access marks a pivotal shift from a vulnerable, network- centric security model to a resilient, identity-centric Zero Trust architecture. By enforcing least- privilege access and rendering private applications invisible to the internet, this project effectively neutralizes the threats of lateral movement and network-based attacks. The solution not only fortifies the organization's security posture against modern cyber threats but also delivers a seamless and high-performance experience for a distributed workforce. Furthermore, the decoupling of access from the network simplifies IT operations, reduces dependency on legacy VPN infrastructure, and provides granular visibility into user-application interactions. Ultimately, this initiative establishes a future-ready foundation for secure digital transformation..

Future Scope:

Stage 1: future scope of web application testing

The future scope of web application testing is promising and evolving rapidly, driven by the continuous growth of the internet and web-based applications. Here are some key aspects of its future scope:

▪ Increased Demand: As more businesses and services move online, the demand for web applications continues to grow. This, in turn, increases the demand for web application testing to ensure security, reliability, and functionality.

▪ Advanced Technologies: The complexity of web applications is rising with the integration of advanced technologies like AI, IoT, and blockchain. Testing these cutting-edge applications will require specialized skills and tools.

▪ Security Emphasis: With the growing number of cyber threats and data breaches, security testing for web applications will become even more critical. Testers will need to focus on vulnerabilities, data protection, and compliance with regulations like GDPR.

▪ Mobile and Cross-Browser Testing: Mobile usage is on the rise, and web applications must be responsive and functional on various devices and browsers. Testing for cross-compatibility will be essential.

▪ Performance Testing: As user expectations for speed and performance increase, load testing, stress testing, and other performance testing types will be crucial to ensure web applications can handle traffic spikes.

Stage 2: future scope of the testing process you understood.

The future scope of the web application testing process, particularly in terms of identifying vulnerabilities, is exceedingly promising due to the continuous growth of digital platforms, the escalating importance of cybersecurity, and the ever-evolving threat landscape. Here are some key aspects that highlight the prospective scope of web application testing for vulnerability discovery:

▪ AI-Driven Testing: The integration of artificial intelligence (AI) and machine learning (ML) into web application testing tools will revolutionize the way vulnerabilities are identified. AI algorithms can swiftly scan vast amounts of code and data to pinpoint vulnerabilities that might elude human testers.

▪ IoT and Mobile App Testing: With the proliferation of Internet of Things (IoT) devices and the increasing usage of mobile applications, the scope of web application testing will extend to encompass these areas. Ensuring the security of IoT devices and the interfaces they use will become critical.

▪ Blockchain and Cryptocurrency Security: As blockchain technology and cryptocurrencies continue to gain traction, security testing will be imperative to safeguard transactions and digital assets. Ensuring the robustness of blockchain-based applications will be a specialized niche in web application testing. Serverless Architectures: With the adoption of serverless computing, web application testing will need to adapt to assess the security of serverless functions and APIs. Ensuring that serverless components are free from vulnerabilities will be paramount. ▪ Container Security: As containerization technologies like Docker and Kubernetes become ubiquitous, web application testing will extend to cover container security. This includes ensuring that container images and orchestration configurations are free from vulnerabilities.

Stage 3: future scope of Aware Access Control

The successful deployment of ZPA lays the groundwork for a broader, more integrated security ecosystem. Future enhancements can be explored to further mature the organization's security capabilities:

▪ Integration with Zscaler Internet Access (ZIA): Converging Private Access (ZPA) with Internet Access (ZIA) to create a unified Secure Access Service Edge (SASE) platform, managing all web and private traffic through a single policy framework.

▪ Advanced Device Posture Checking: Moving beyond basic compliance to incorporate realtime risk scores from EDR tools, blocking access if a device shows signs of active compromise.

▪ Privileged Remote Access: Extending ZPA to third-party vendors and contractors with time-bound, audited access to specific critical systems, replacing jump boxes and bastion hosts.

▪ IoT/OT Segmentation: Applying the same micro-tunneling principles to Internet of Things (IoT) and Operational Technology (OT) devices to ensure they can only communicate with specific management servers.

▪ AI-Driven Anomaly Detection: Leveraging machine learning on ZPA logs to detect behavioral anomalies, such as a user accessing an application at an unusual hour or from a strange location, triggering automated MFA challenges or access revocation.

▪ Cloud Workload Protection: Utilizing ZPA to secure connections between containers and microservices in multi-cloud environments (AWS, Azure, GCP), ensuring east-west traffic between workloads is encrypted and authenticated.

▪ Automated Provisioning/Deprovisioning: Enhancing the identity provider integration to automatically revoke application access the moment an employee is terminated or changes roles, ensuring near real-time access control.

▪ Digital Experience Monitoring (DEM): Implementing DEM tools to proactively monitor the end- user experience, identifying network latency or application performance issues before they impact productivity.

Topics Explored:

1. Zscaler
2. Zero trust
3. ZPA
4. VPN
5. MISP
6. College Network Information
7. Deployment of SOC in the College
8. Threat Intelligence
9. Incident Response

Tools Explored:

1. Nessus Scanner